Lecture 6: Introduction to quantum mechanics-2

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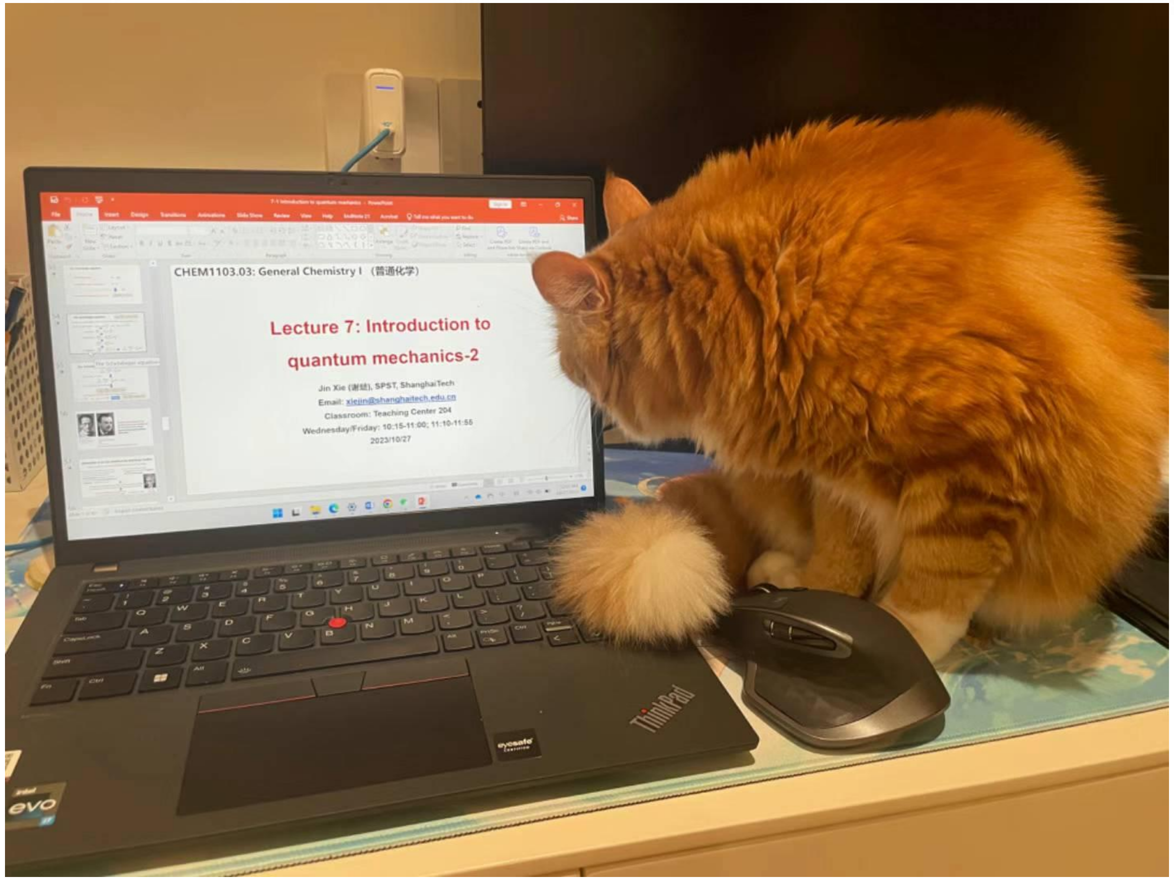
Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/10/17

Mid-term 1

- When: 2025/10/29, 90 minutes (subject to change)
- What: introduction – quantum mechanics (subject to change)



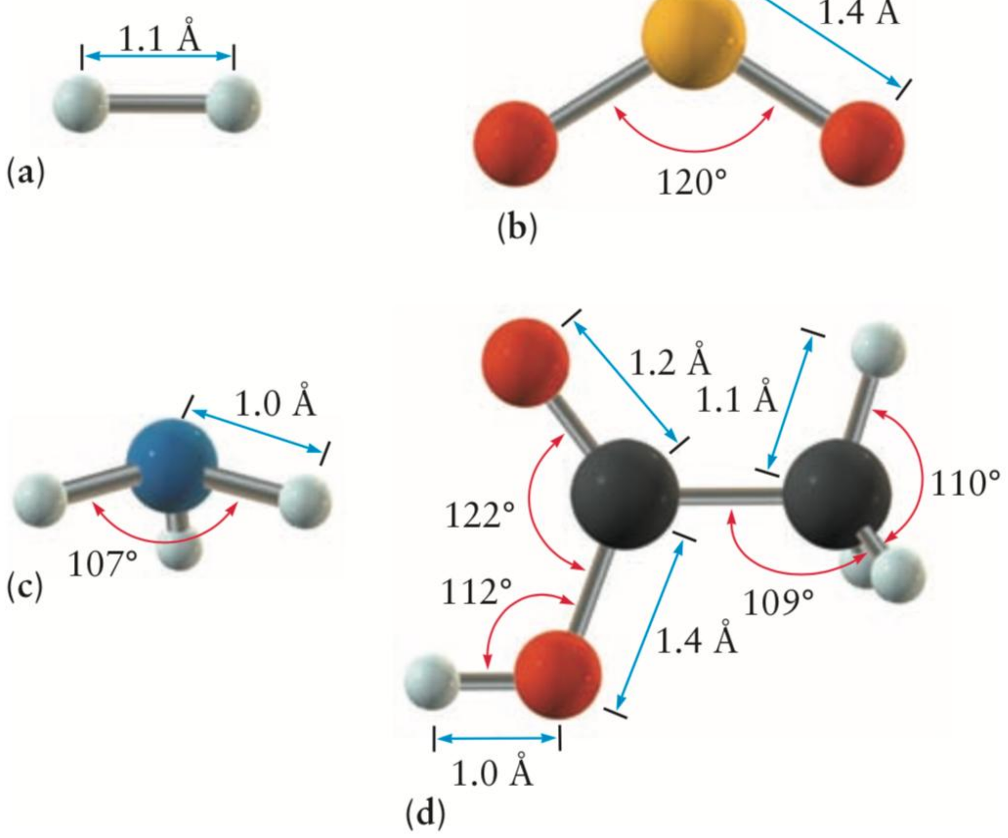
Major points of last lecture

- **Atomic shells and classical models of chemical bonding**

  1. **The shapes of molecules**
  2. **Oxidation numbers**
  3. **Inorganic nomenclature**

The shapes of molecules

- **Bond length**: distance between the atomic nuclei in a particular bond
- **Bond angle**: angle between the axes of adjacent bonds
- **Dihedral angle**: angle between two planes, each defined by three atoms with the planes having two atoms in common



Molecular shape or geometry is governed by energetics; a molecule assumes the geometry that gives it the lowest potential energy.

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

- **The Valence Shell Electron-Pair Repulsion Theory (VSEPR): 价层电子对互斥模型**

⬇

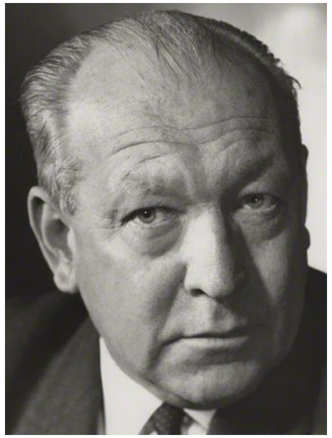
- **fundamental idea: the electron pairs position themselves as far apart as possible to minimize their repulsions**

⬇

$$SN = \begin{pmatrix}\text{number of atoms} \\ \text{bonded to central atom}\end{pmatrix} + \begin{pmatrix}\text{number of lone pairs} \\ \text{on central atom}\end{pmatrix}$$

SN: steric number (空间位数)

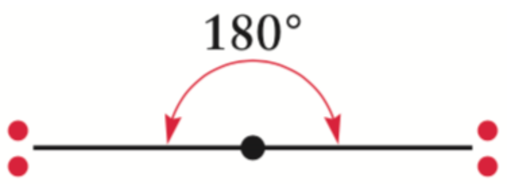
Ronald J. Gillespie (1924–)



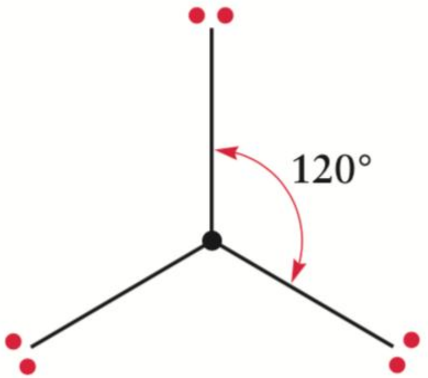
Ronald S. Nyholm (1917–1971)

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

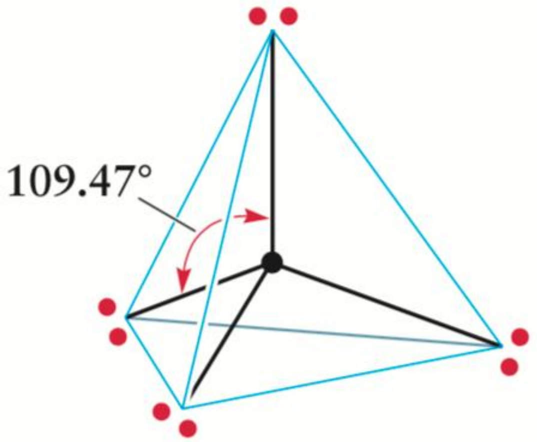
SN: steric number (空间位数)



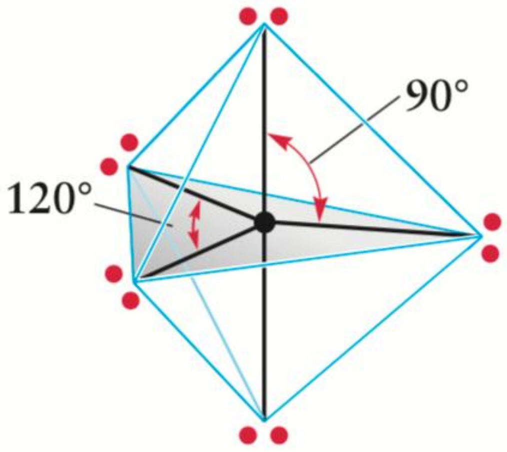
2: Linear



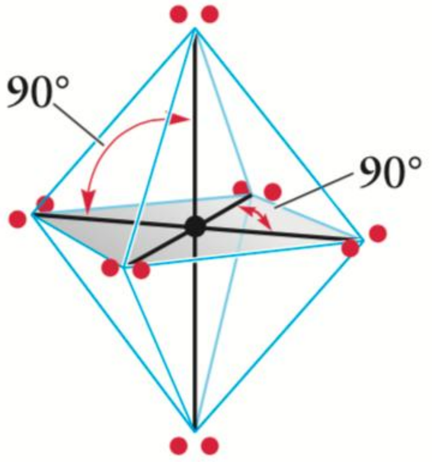
3: Trigonal planar



4: Tetrahedral



5: Trigonal bipyramidal



6: Octahedral

The Valence Shell Electron-Pair Repulsion Theory (VSEPR)

Step 1: Rewrite a structure as AX_mE_n, m = number of ligands, n = number of lone pairs

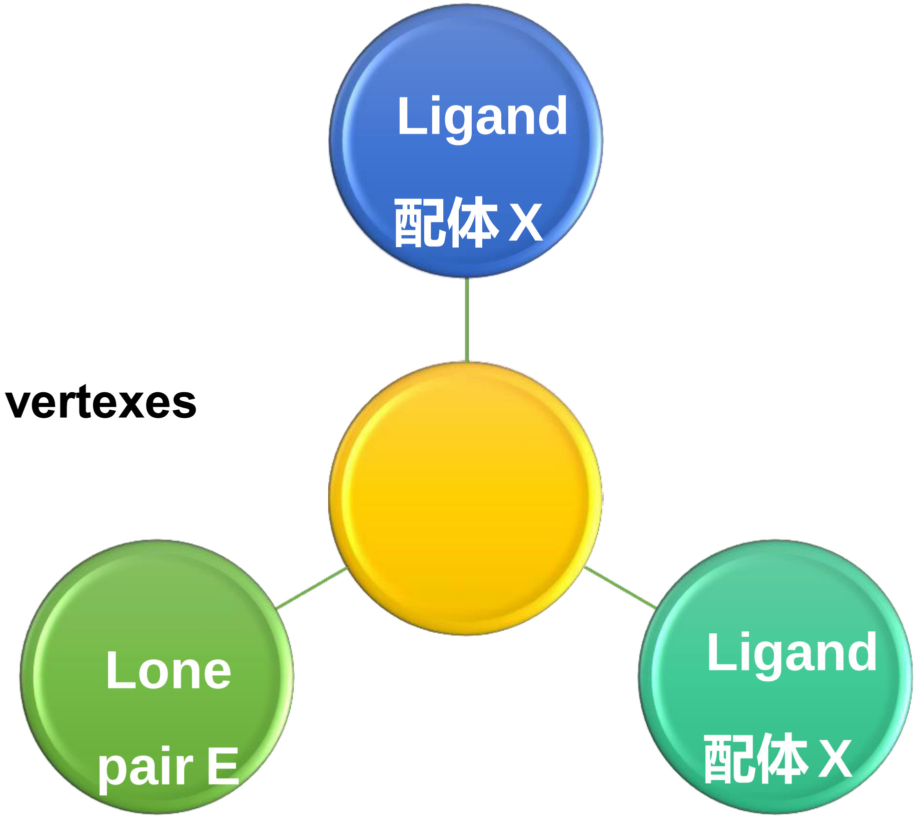
Step 2: Assign a Steric Number; SN = m + n

Step 3: Place the ligands and lone pairs on a polyhedron of SN vertexes

Step 4: Repulsion: E-E > E-X > X-X;

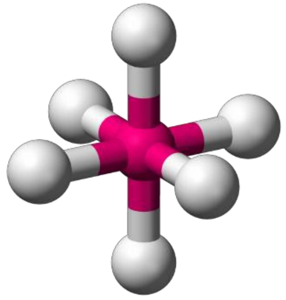
triple bond>double bond>single bond

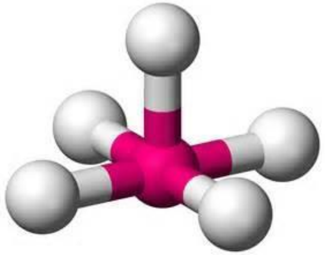
X high electronegativity (EN), less repulsion

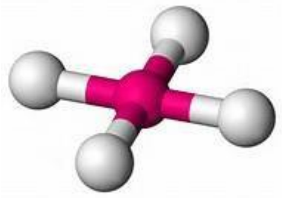


Summary of molecular geometry

SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX_2	Linear (线形）
3	AX_3	Trigonal planar （三角形）
	AX_2E	Bent （弯曲形）
4	AX_4	Tetrahedral （四面体）
	AX_3E	Trigonal pyramidal （三角锥）
	AX_2E_2	Bent
5	AX_5	Trigonal bipyramid （三角双锥）
	AX_4E	Distorted see-saw （跷跷板）
	AX_3E_2	Distorted T （T字）
	AX_2E_3	Linear
6	AX_6	Octahedral （八面体）
	AX_5E	Square pyramidal （四角锥）
	AX_4E_2	Square planar （正方平面）







Major points of last lecture

- Introduction to quantum mechanics

1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. The Bohr model: predicting discrete energy levels in atoms

The word quantum is the neuter singular of the Latin interrogative adjective quantus, meaning "how much".

quantum

noun [C] • PHYSICS • specialized

UK /ˈkwɒn.təm/ **US** /ˈkwɑːn.t̬əm/

plural **quanta**

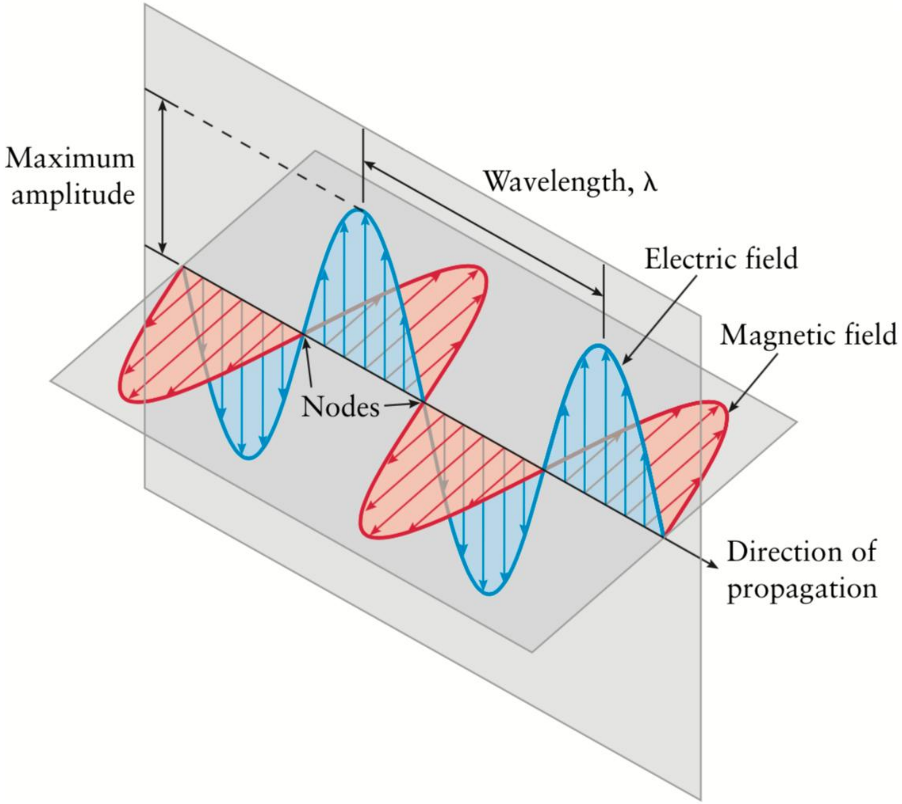
the smallest amount or unit of something, especially energy:

mechanics [U]

the study of the effect of physical forces on objects and their movement

Preliminaries: light (electromagnetic radiation?)

Name	Integral equations	Differential equations
Gauss's law	$\oiint_{\partial\Omega} \mathbf{E}\cdot\mathrm{d}\mathbf{S} = \frac{1}{\varepsilon_0}\iiint_{\Omega}\rho\,\mathrm{d}V$	$\nabla\cdot\mathbf{E} = \frac{\rho}{\varepsilon_0}$
Gauss's law for magnetism	$\oiint_{\partial\Omega} \mathbf{B}\cdot\mathrm{d}\mathbf{S} = 0$	$\nabla\cdot\mathbf{B} = 0$
Maxwell–Faraday equation (Faraday's law of induction)	$\oint_{\partial\Sigma} \mathbf{E}\cdot\mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}}{\mathrm{d}t}\iint_{\Sigma}\mathbf{B}\cdot\mathrm{d}\mathbf{S}$	$\nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t}$
Ampère's circuital law (with Maxwell's addition)	$\oint_{\partial\Sigma} \mathbf{B}\cdot\mathrm{d}\boldsymbol{l} = \mu_0\left(\iint_{\Sigma}\mathbf{J}\cdot\mathrm{d}\mathbf{S} + \varepsilon_0\frac{\mathrm{d}}{\mathrm{d}t}\iint_{\Sigma}\mathbf{E}\cdot\mathrm{d}\mathbf{S}\right)$	$\nabla\times\mathbf{B} = \mu_0\left(\mathbf{J} + \varepsilon_0\frac{\partial\mathbf{E}}{\partial t}\right)$

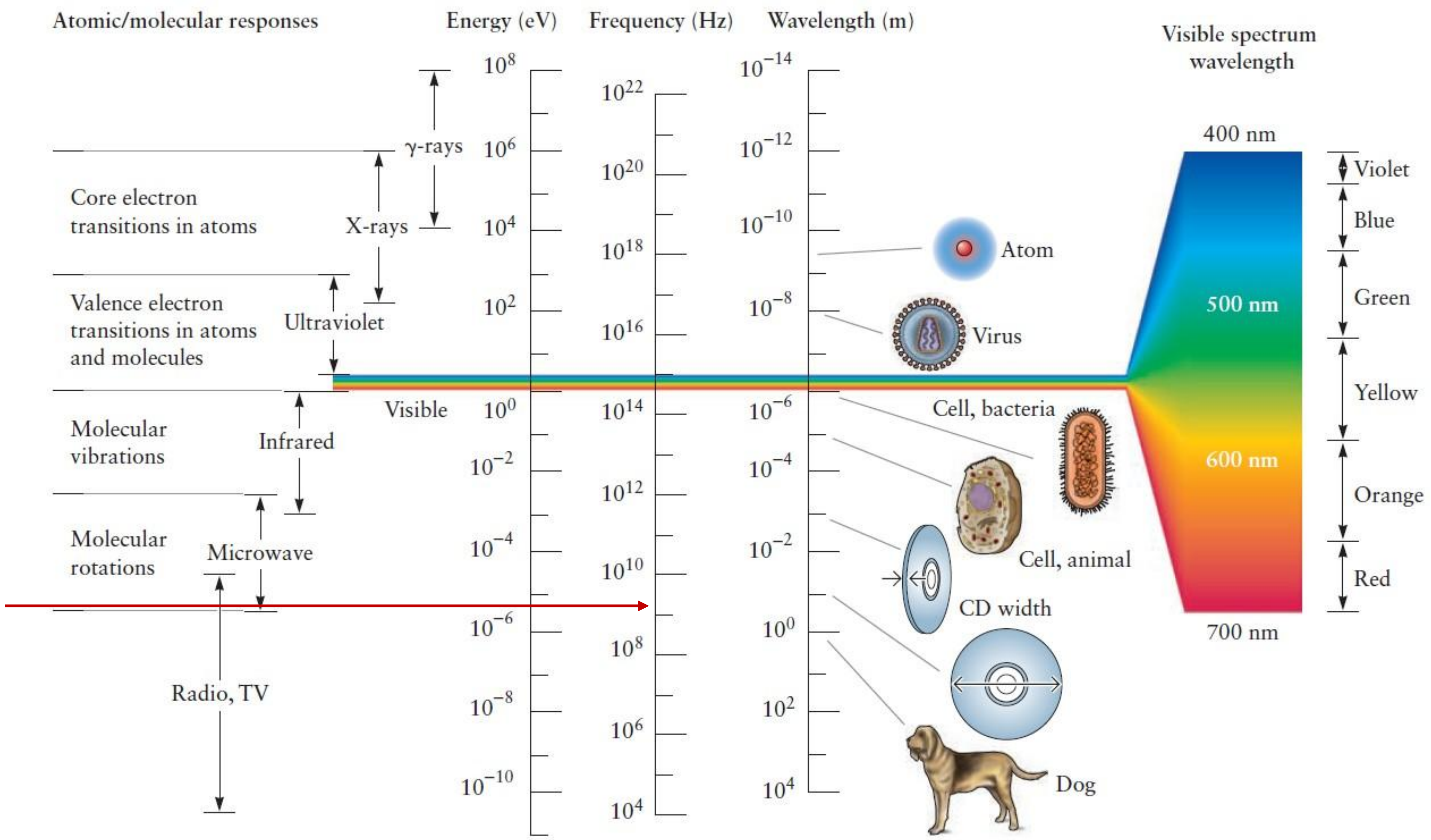


- **Predicted the existence of electromagnetic radiation.**
- **Predicted visible light is a form of electromagnetic radiation.**

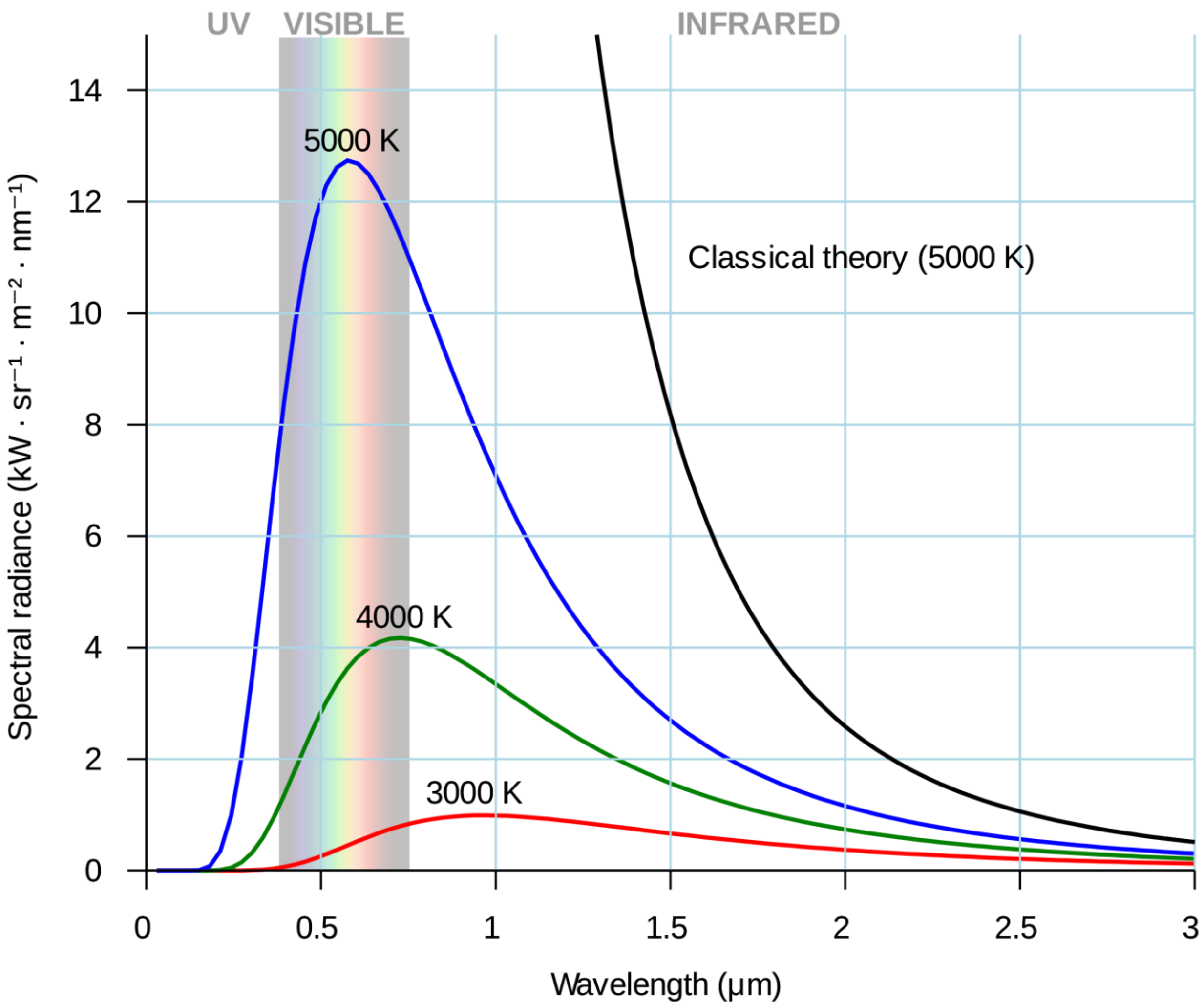
Visible light as a propagating wave of electromagnetic carries both energy and momentum.

James Clerk Maxwell (1831-1879)

Preliminaries: light (electromagnetic radiation?)



Blackbody Radiation



Blackbody: an idealized physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence.

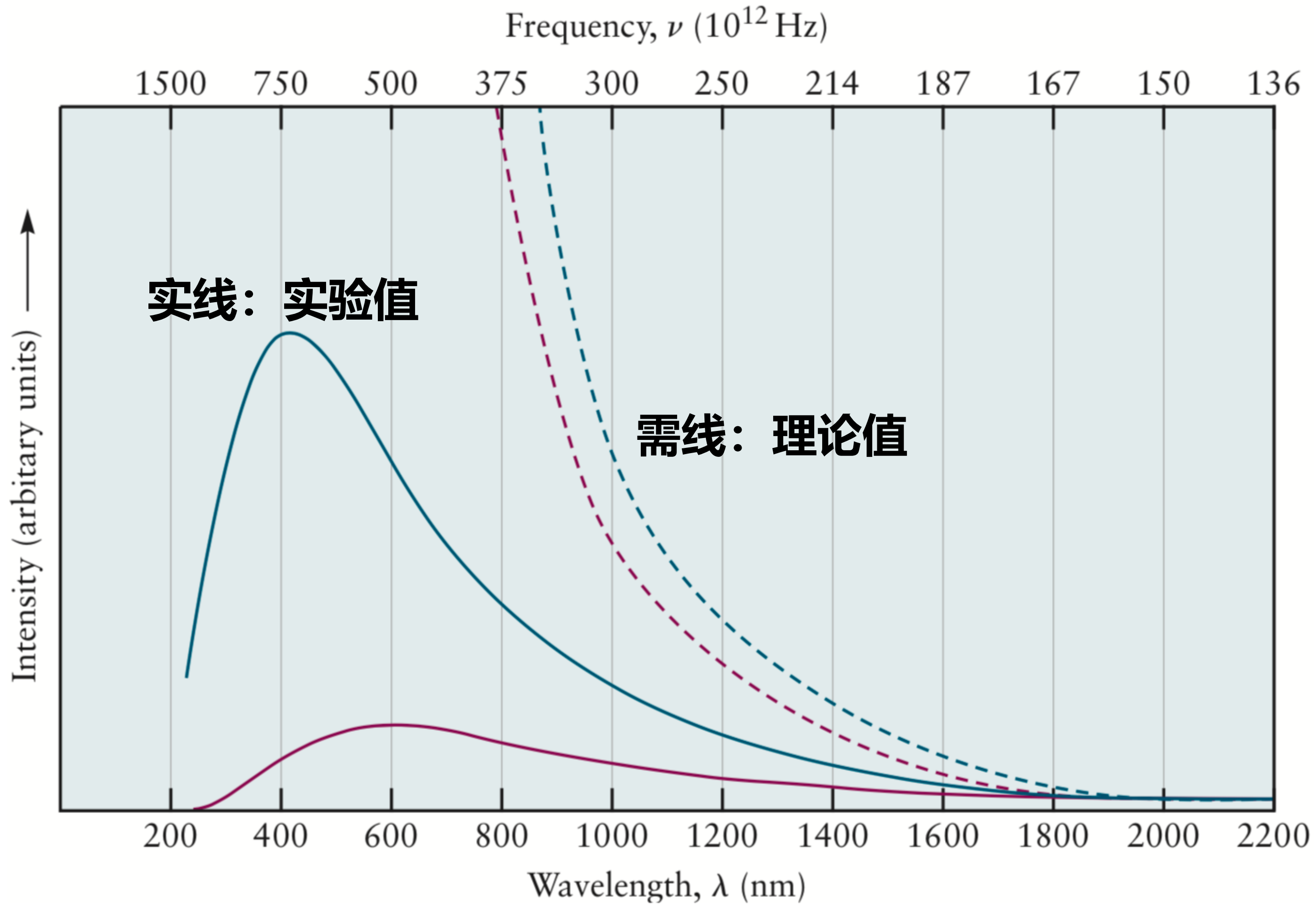
A blackbody is not necessary black in color.

A black body in thermal equilibrium (that is, at a constant temperature) emits electromagnetic radiation called blackbody radiation. It has a spectrum that is determined by the temperature alone, not by the body's shape or composition.

Blackbody Radiation

Classical physics:

$$\rho_T(\nu) = \frac{8\pi k_B T \nu^2}{c^3}$$



Rayleigh-Jeans law

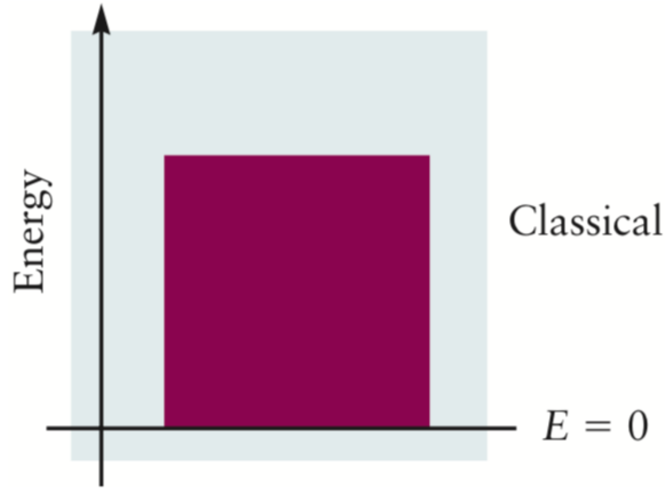
- $\rho_T(\nu)$ is the intensity of the radiation at the frequency ν;
- k_B is a fundamental constant called the Boltzmann constant
- T is the temperature in kelvins (K)
- c is the speed of light.

"ultraviolet catastrophe": predicts an infinite intensity at very short wavelengths, whereas the experimental intensities always remain finite and actually fall to zero at very short wavelengths.

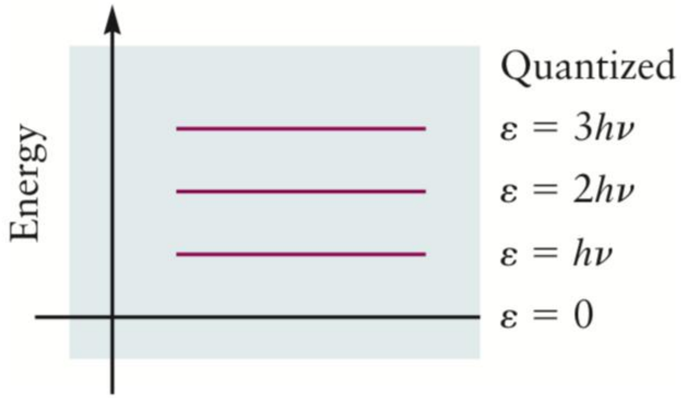
Blackbody Radiation

Planck's 1st hypothesis: It is not possible to put an arbitrary amount of energy into an oscillator of frequency. Instead, the oscillator must gain and lose energy in "packets," or quanta, of magnitude hν. The total energy of an oscillator can take only discrete values that are integral multiples of hν.

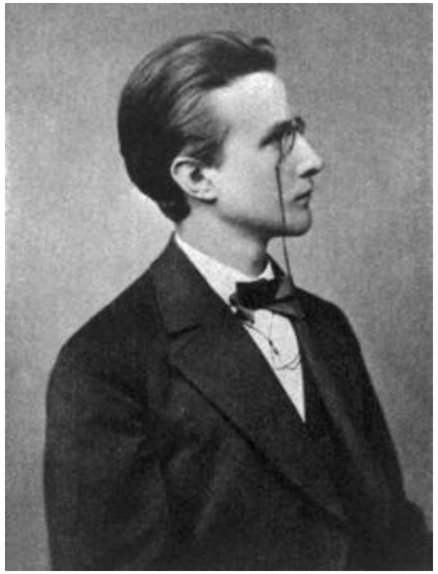
$$\varepsilon_{osc} = nh\nu \qquad n = 1, 2, 3, 4, \ldots$$



An oscillator obeying classical mechanics has continuous values of energy and can gain or lose energy in arbitrary amounts.



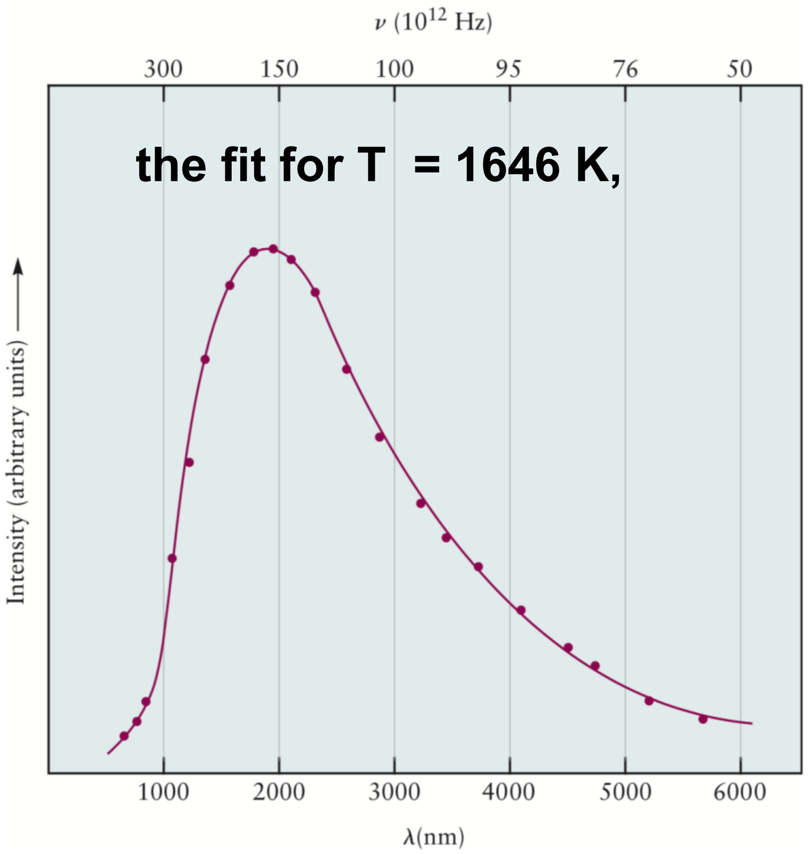
An oscillator described by Planck's postulate has discrete energy levels. It can gain or lose energy only in amounts that correspond to the difference between two energy levels.



Max Planck (1858-1947)

Blackbody Radiation

Planck's 2^{nd} step: the excitation of a particular oscillator is an all-or-nothing event. The falloff in intensity with frequency at a given temperature of the blackbody radiation is due to a diminishing probability of exciting the high-frequency oscillators.



$$\rho_T(\nu) = \frac{8\pi h\nu^3}{c^3}\,\frac{1}{e^{h\nu/k_B T} - 1}$$

Max Planck
(1858–1947)

Planck's constant

In the revised SI, the Planck constant h is equal to exactly

6.626 070 15 × 10^{-34} Joule seconds

Blackbody Radiation

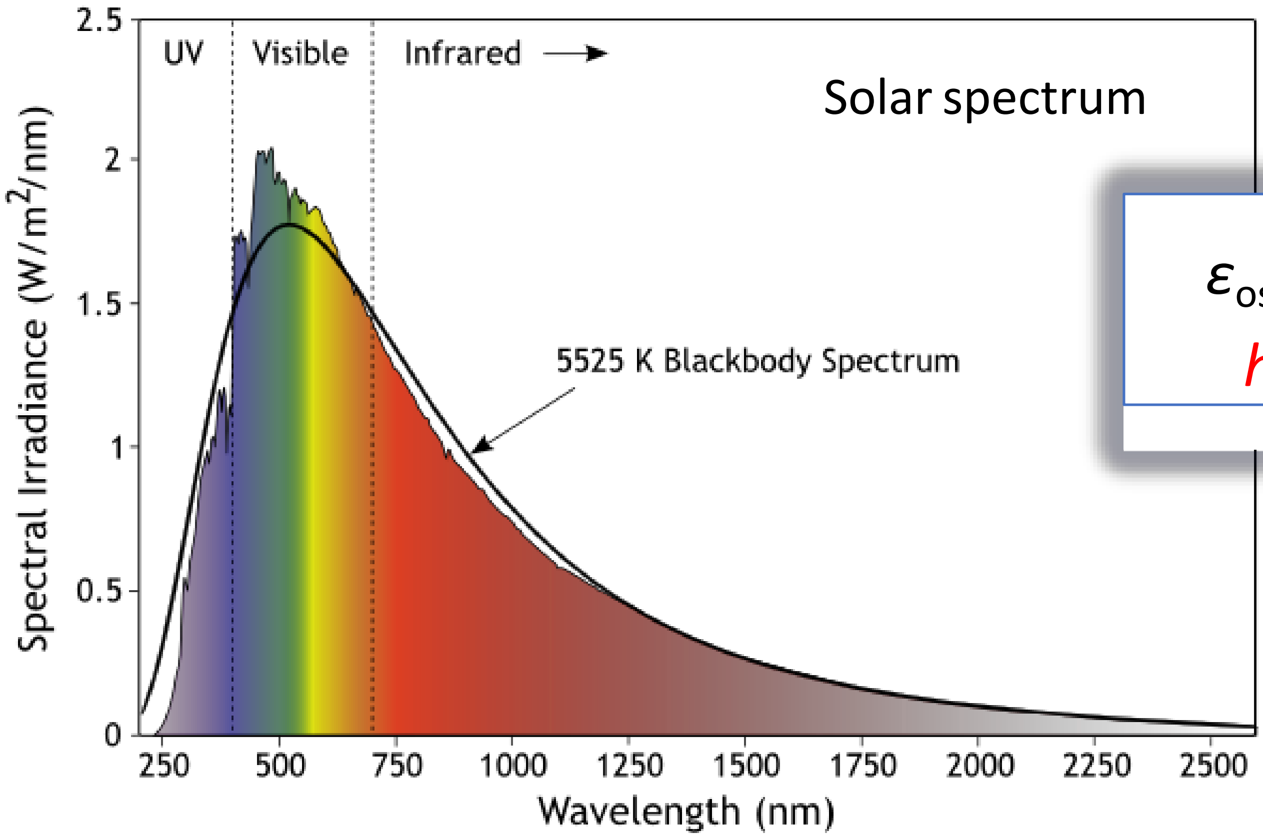
Planck's dramatic explanation of blackbody radiation includes three fundamentally new ideas:

- **The energy of a system can take only discrete values, which are represented on its energy-level diagram.**

- **A quantized oscillator can gain or lose energy only in discrete amounts $\Delta\varepsilon$, which are related to its frequency by $\Delta\varepsilon = h\nu$.**

- **To emit energy from higher energy states, the temperature of a quantized system must be sufficiently high to excite those states.**

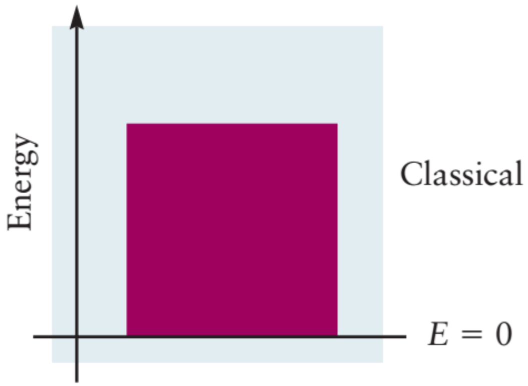
Blackbody Radiation

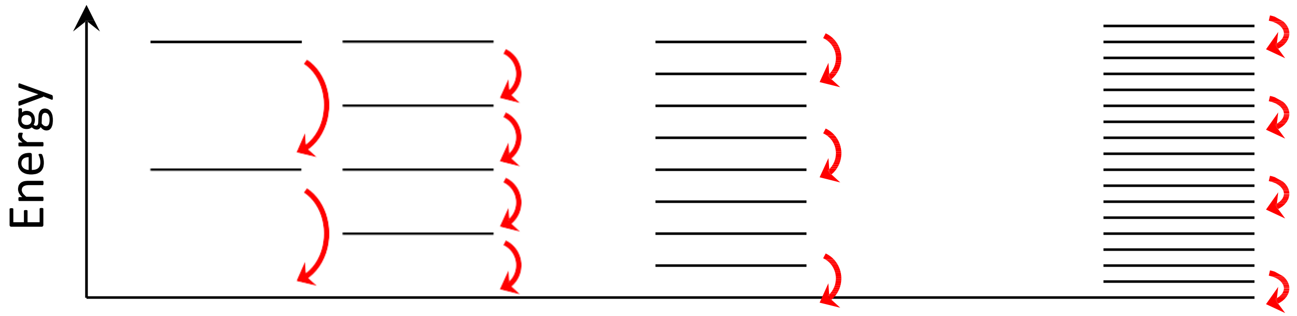


$$\nu = \frac{c}{\lambda}$$

$$\varepsilon_{osc} = E = nh\nu,\ n \in \mathbb{N}$$

$$h = 6.63\times10^{-34}\ \text{J}\cdot\text{s}$$





$h\nu_1$

$h\nu_2$

$h\nu_3$

Atomic spectra

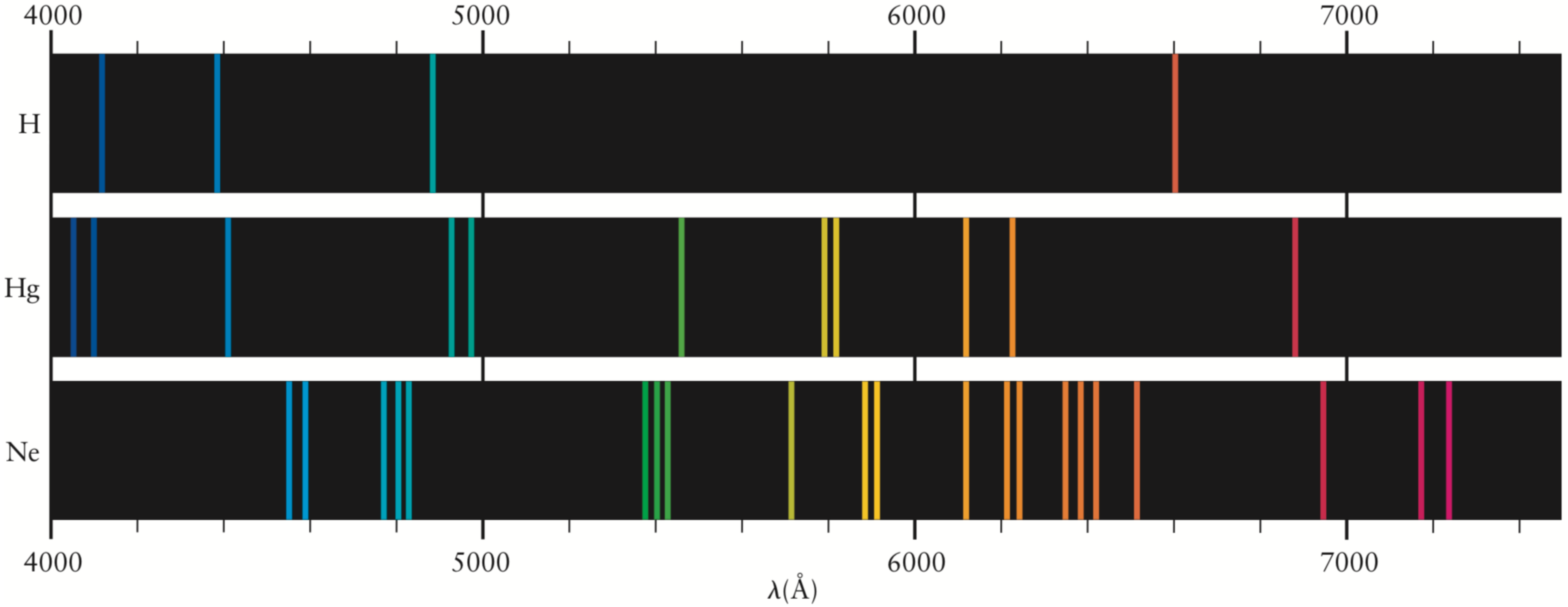
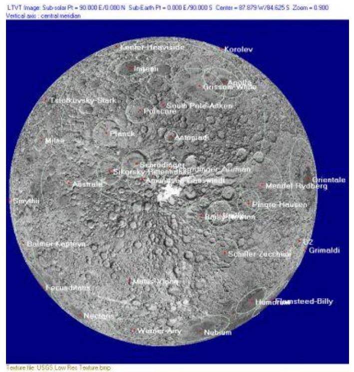


FIGURE 4.10 Atoms of hydrogen, mercury, and neon emit light at discrete wavelengths. The pattern seen is characteristic of the element under study. 1 Å = 10^{-10} m.

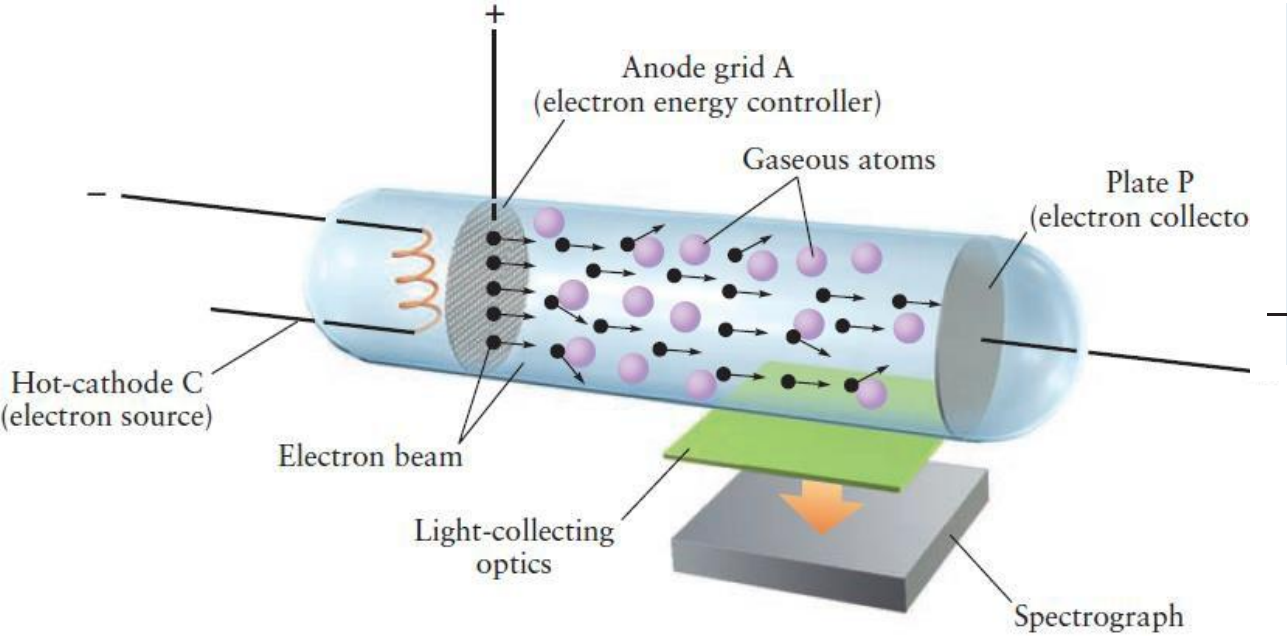
1885, J. J. Balmer discovered that hydrogen atoms emit a series of lines in the visible region, with frequencies given by the following simple formula:

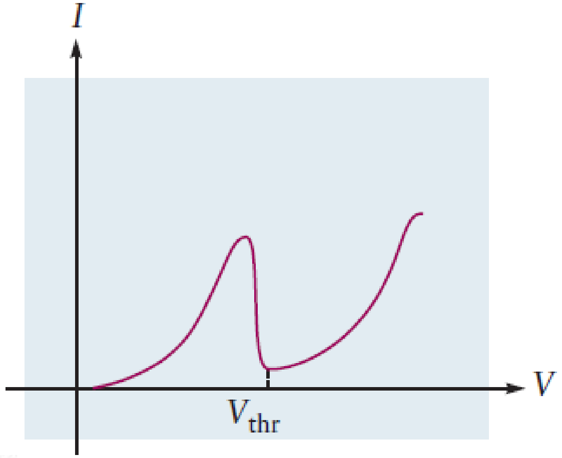
$$\nu = \left[\frac{1}{4} - \frac{1}{n^2}\right] \times 3.29 \times 10^{15}\ \mathrm{s}^{-1} \qquad n = 3,\ 4,\ 5, \ldots$$

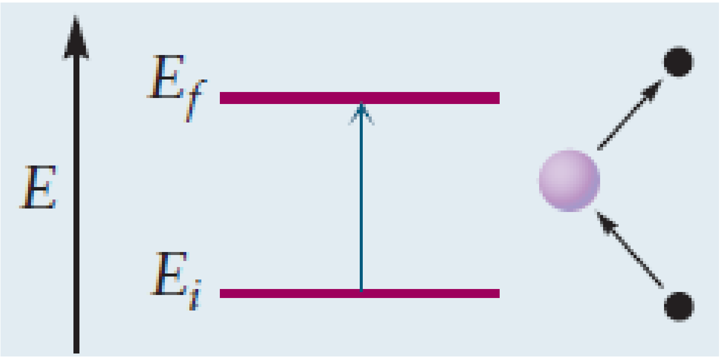


The Franck–Hertz Experiment

Test Bohr's hypothesis that the energy of atoms is quantized







The current shows a sharp change at a particular value of the accelerating voltage, corresponding to the threshold energy transfer from the electron to a gaseous atom.

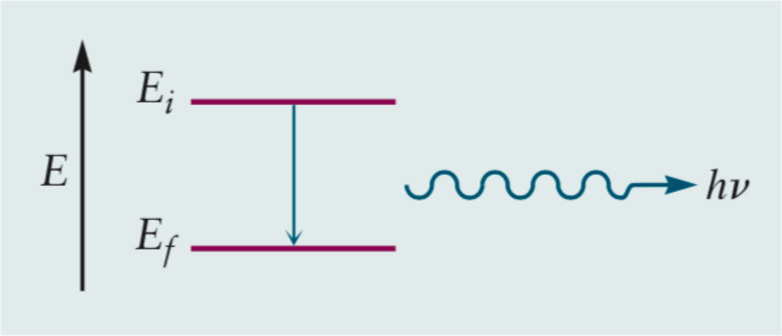
When the accelerating voltage was slightly above V_{thr}, a single emission line was observed whose frequency was very nearly equal to

$$\nu = \frac{\Delta E}{h} = \frac{eV_{thr}}{h}$$

Atomic spectra

Niels Bohr proposed a model:

1. **the hydrogen atom allowed only discrete energy states to exist;**
2. **light absorption resulted from a transition of the atoms between two of these states.**



$$\nu = \frac{E_f - E_i}{h} \qquad \text{or} \qquad \Delta E = h\nu$$

h is Planck's constant

Niels Bohr

(1885-1962)

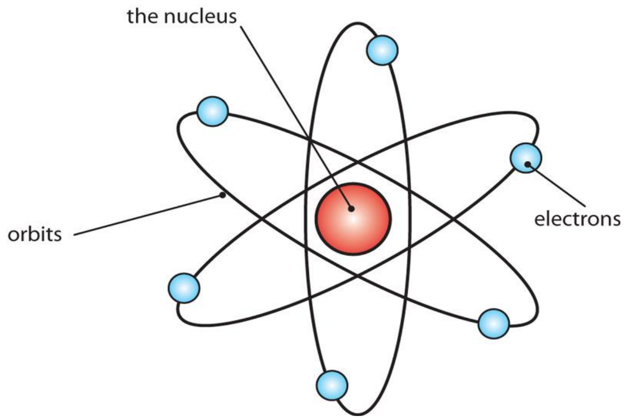
- **In an atom, there are various states 能态 of fixed, discrete energy.**
- **Emission or absorption of light results from a transition 跃迁 between two of these states.**
- **The energy quantum *hv* of the absorbed/emitted light equals the transition energy.**

Major points of today's lecture

- **Introduction to quantum mechanics**

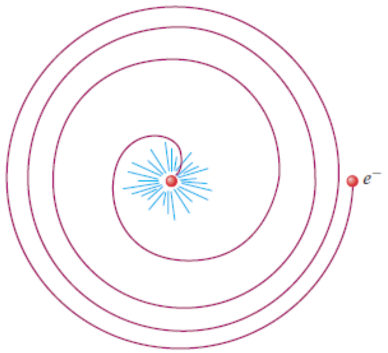
1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. The Bohr model: predicting discrete energy levels in atoms
4. Evidence for wave-particle duality
5. The Schrödinger equation
6. Quantum mechanics of particle-in-a-box models
7. Wave functions for particles in two- and three-dimensional boxes

The Bohr model: predicting discrete energy levels in atoms



Rutherford Model

(1911)



Classical theory states that atoms constructed according to Rutherford's nuclear model are not stable. The motion of electrons around the nucleus would cause them to radiate energy and quickly spiral into the nucleus.

The total energy of the hydrogen atom:

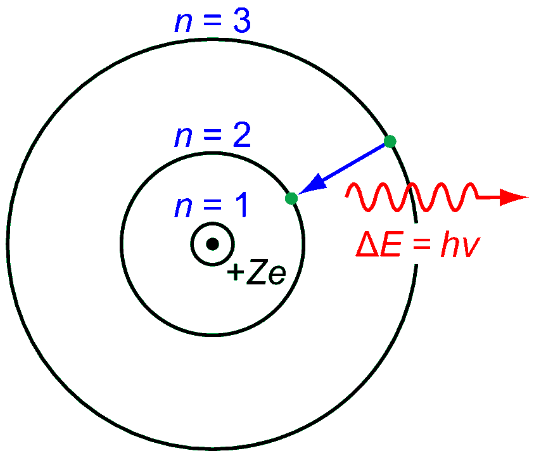
$$E = \frac{1}{2} m_e v^2 - \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$|F_{\text{Coulomb}}| = |m_e a|$$

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m_e \frac{v^2}{r}$$

The Bohr model: predicting discrete energy levels in atoms

Bohr Model



Bohr model:

- **only certain discrete orbits are allowed**

  (characterized by radius r_n and energy E_n)
- **light is emitted or absorbed only when the electron "jumps" from one stable orbit to another**

Bohr postulate: the angular momentum is quantized in integral multiples of h/2π

- **Linear momentum: $p=m_e v$**
- **Angular momentum: $L=m_e vr$**

$$L = m_e vr = n\frac{h}{2\pi} \qquad n = 1, 2, 3, \ldots$$

The Bohr model: predicting discrete energy levels in atoms

$$L = m_e vr = n\frac{h}{2\pi} \qquad n = 1, 2, 3, \ldots$$

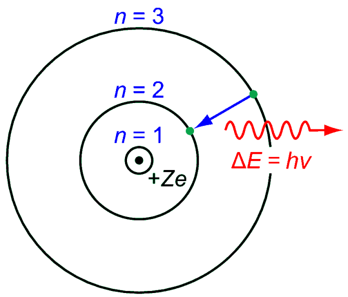
$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m_e\frac{v^2}{r} \quad \Rightarrow \quad r_n = \frac{\epsilon_0 n^2 h^2}{\pi Z e^2 m_e} = \frac{n^2}{Z}a_0$$

a_0, the Bohr radius, has the numerical value 5.29×10^{-11} m = 0.529 Å. $(a_0 = \epsilon_0 h^2/\pi e^2 m_e)$

The Bohr model: predicting discrete energy levels in atoms

1st prediction: the existence of a series of orbits whose distances from the nucleus increase dramatically with increasing n.

$$r_n = \frac{\epsilon_0 n^2 h^2}{\pi Z e^2 m_e} = \frac{n^2}{Z} a_0$$



$$L = m_e v r = n \frac{h}{2\pi} \qquad n = 1, 2, 3, \ldots$$

The velocity v_n corresponding to the orbit with radius r_n.

$$v_n = \frac{nh}{2\pi m_e r_n} = \frac{Ze^2}{2\epsilon_0 nh}$$

$$E = \frac{1}{2} m_e v^2 - \frac{Ze^2}{4\pi\epsilon_0 r}$$

2nd prediction: a discrete energy-level diagram for the one-electron atom. The ground state (基态) is identified by n=1, and the excited states (激发态) have higher values of n.

$$E_n = \frac{-Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} = -(2.18 \times 10^{-18}\ \mathrm{J}) \frac{Z^2}{n^2} \qquad n = 1, 2, 3, \ldots$$

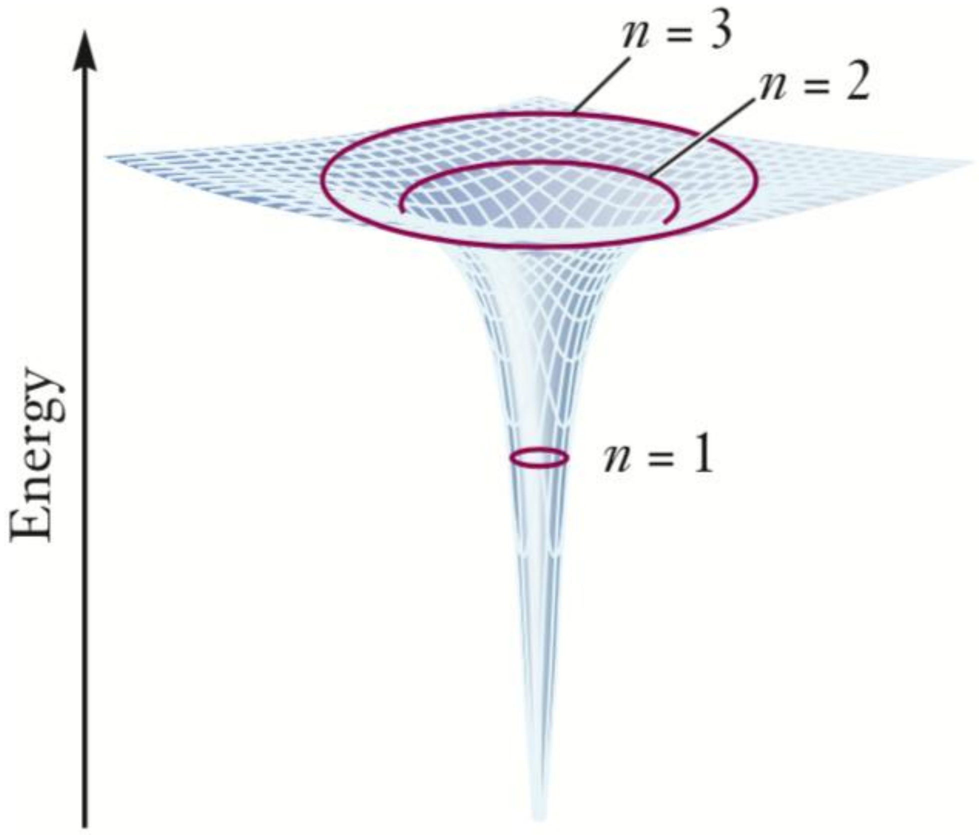
The Bohr model: predicting discrete energy levels in atoms

Allow values of the energy

$$E_n = \frac{-Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} = -(2.18 \times 10^{-18}\,\mathrm{J})\,\frac{Z^2}{n^2} \qquad n = 1, 2, 3, \ldots$$

$$E_n = -\frac{Z^2}{n^2} \quad \text{(rydberg)} \qquad n = 1, 2, 3, \ldots$$

1 rydberg = 2.18 × 10^{-18} J



The ionization energy:

$$\Delta E = E_{\text{final}} - E_{\text{initial}} = 0 - (-2.18 \times 10^{-18}\ \mathrm{J}) = 2.18 \times 10^{-18}\ \mathrm{J}$$

$$IE = (6.022 \times 10^{23}\ \text{atoms mol}^{-1})(2.18 \times 10^{-18}\ \text{J atom}^{-1}) = 1.31 \times 10^{6}\ \text{J mol}^{-1} = 1310\ \text{kJ mol}^{-1}$$

This prediction agrees with the experimentally observed ionization energy of hydrogen atoms and provides confidence in the validity of the Bohr model.

Consider the $n = 2$ state of Li^{2+}. Using the Bohr model, calculate the radius of the electron orbit, the electron velocity, and the energy of the ion relative to that of the nucleus and electron separated by an infinite distance.

Solution

Because $Z = 3$ for Li^{2+} (the nuclear charge is $+3e$) and $n = 2$, the radius is

$$r_2 = \frac{n^2}{Z} a_0 = \frac{4}{3} a_0 = \frac{4}{3}(0.529\ \text{Å}) = 0.705\ \text{Å}$$

The velocity is

$$v_2 = \frac{nh}{2\pi m_e r_2} = \frac{2(6.626 \times 10^{-34}\ \text{J s})}{2\pi(9.11 \times 10^{-31}\ \text{kg})(0.705 \times 10^{-10}\ \text{m})} = 3.28 \times 10^6\ \text{m s}^{-1}$$

The energy is

$$E_2 = -\frac{(3)^2}{(2)^2}(2.18 \times 10^{-18}\ \text{J}) = -4.90 \times 10^{-18}\ \text{J}$$

Typically, atomic sizes fall in the range of angstroms, and atomic excitation energies in the range of 10^{-18} J. This is consistent with the calculations of coulomb potential energies in electron volts and dimensions in angstroms in Section 3.3.

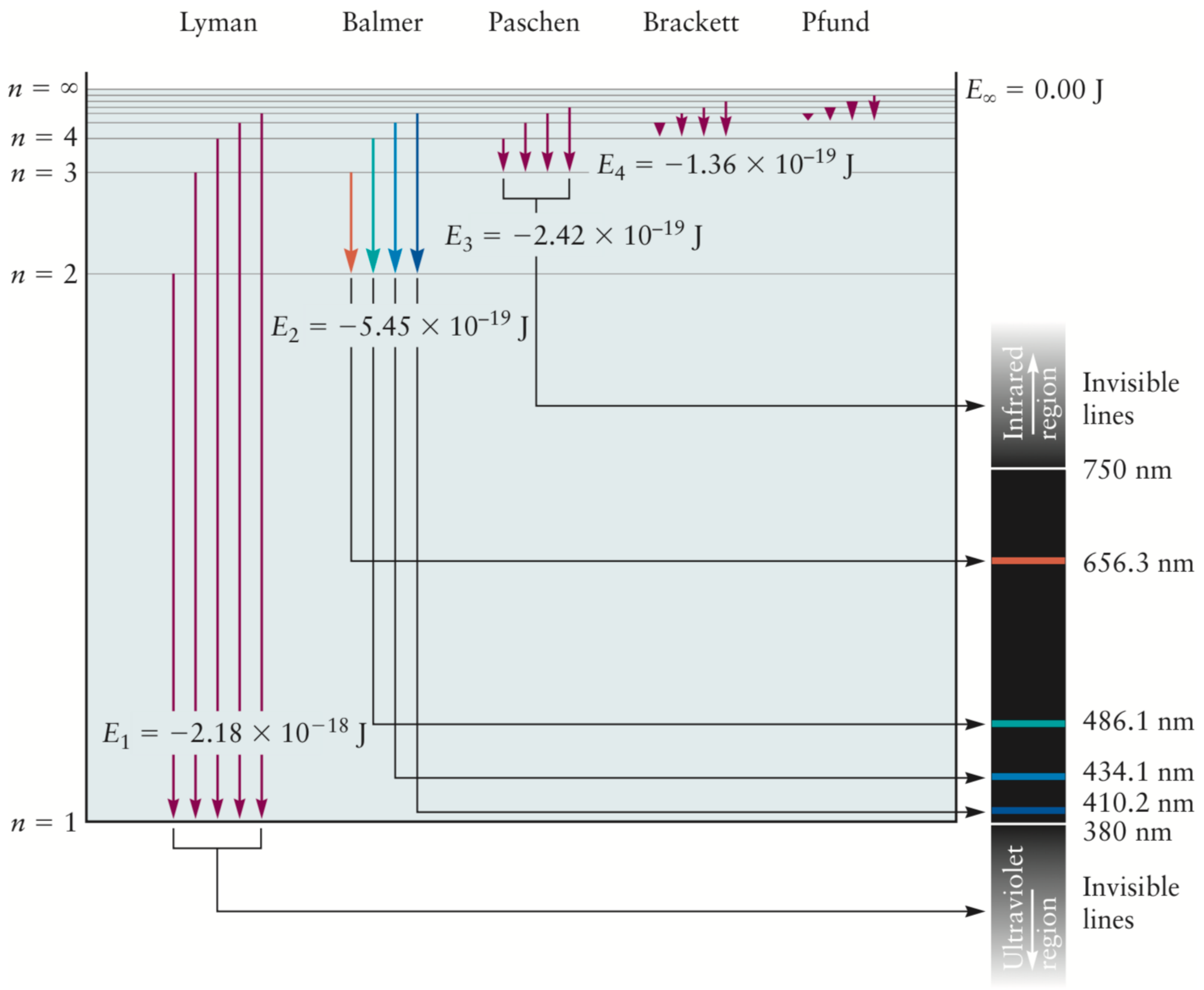
The Bohr model: predicting discrete energy levels in atoms

When a one-electron atom or ion undergoes a transition from a state characterized by quantum number n_i to a state lower in energy with quantum number n_f ($n_i > n_f$), light is emitted to carry off the energy $h\nu$ lost by the atom.

$$h\nu = \frac{Z^2 e^4 m_e}{8\epsilon_0^2 h^2}\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right] \qquad \text{(emission)}$$

$$\nu = \frac{Z^2 e^4 m_e}{8\epsilon_0^2 h^3}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = (3.29 \times 10^{15}\ \mathrm{s}^{-1})\, Z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$n_i > n_f = 1, 2, 3, \ldots$ (emission)

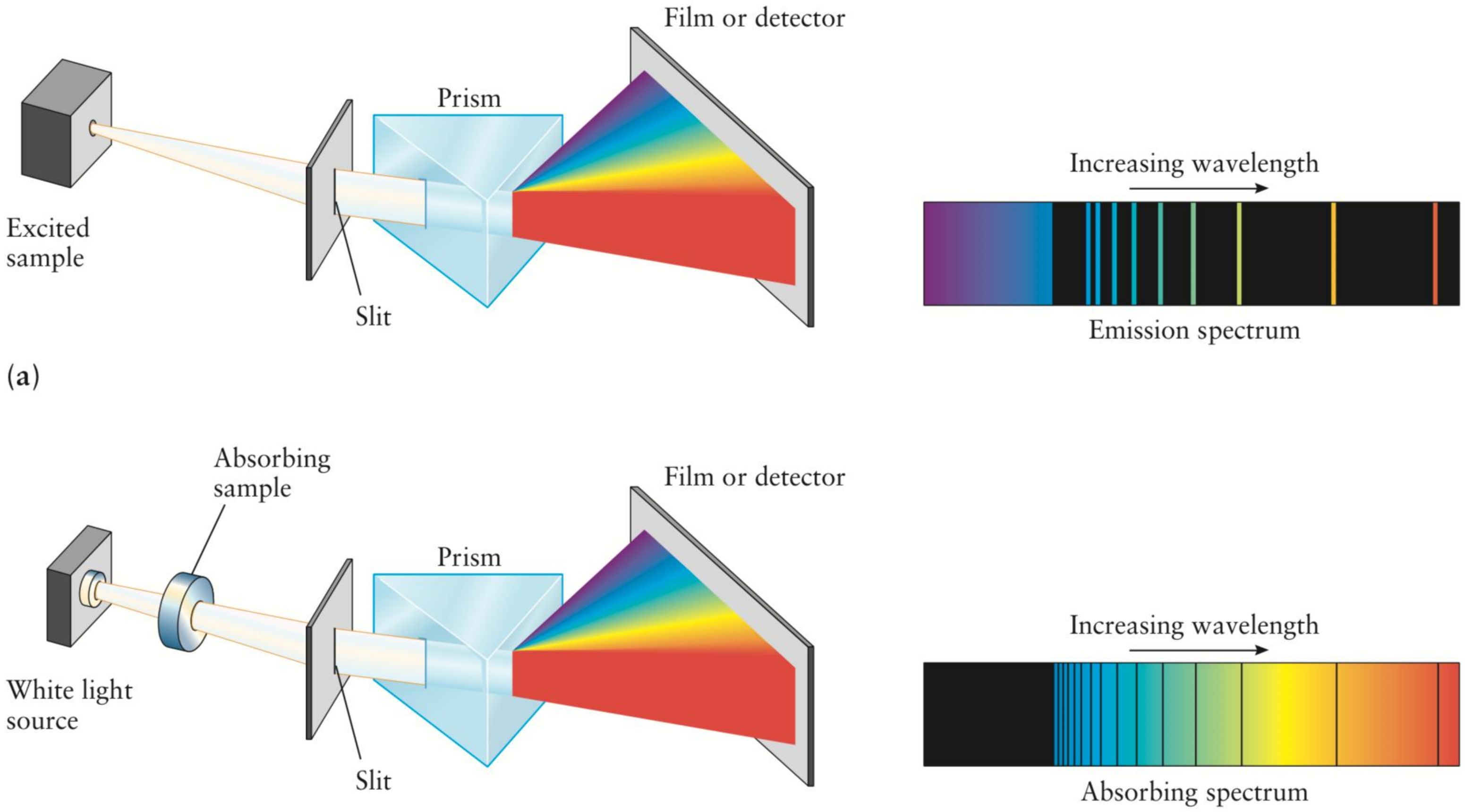


The Bohr model: predicting discrete energy levels in atoms

Conversely, an atom can absorb energy hν from a photon as it undergoes a transition to a higher energy state ($n_f > n_i$). In this case, conservation of energy requires $E_i + h\nu = E_f$; thus, the absorption spectrum shows a series of lines at frequencies.

$$\nu = (3.29 \times 10^{15}\ \mathrm{s}^{-1})\, Z^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

$$n_f > n_i = 1, 2, 3, \ldots \text{ (absorption)}$$



Atomic spectra

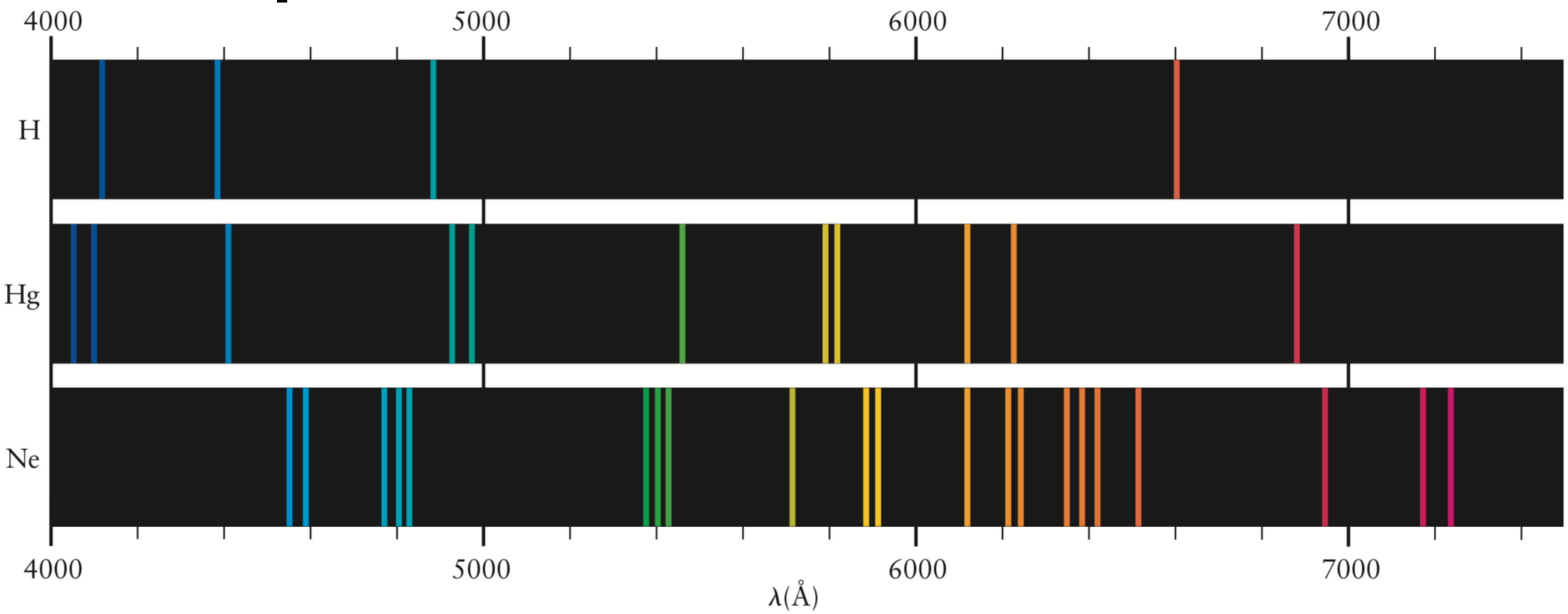
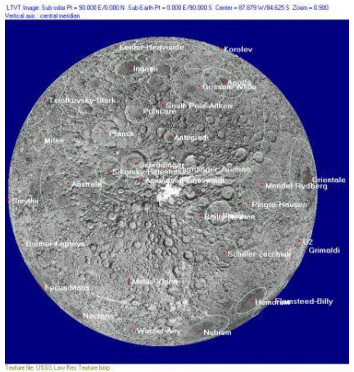


FIGURE 4.10 Atoms of hydrogen, mercury, and neon emit light at discrete wavelengths. The pattern seen is characteristic of the element under study. 1 Å = 10^{-10} m.

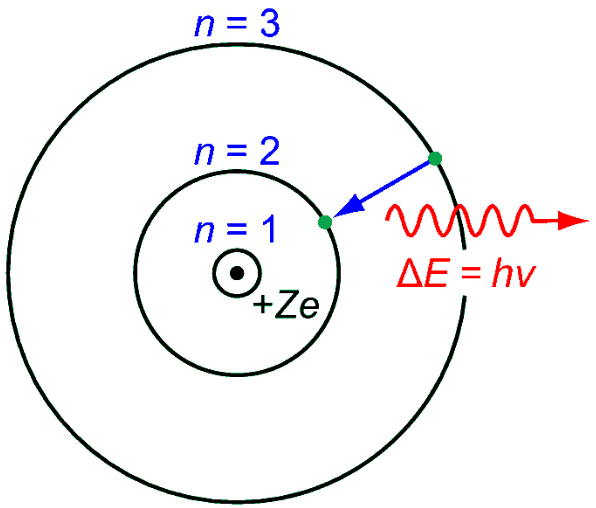
1885, J. J. Balmer discovered that hydrogen atoms emit a series of lines in the visible region, with frequencies given by the following simple formula:

$$\nu = \left[\frac{1}{4} - \frac{1}{n^2}\right] \times 3.29 \times 10^{15}\ \mathrm{s}^{-1} \qquad n = 3,\ 4,\ 5, \ldots$$



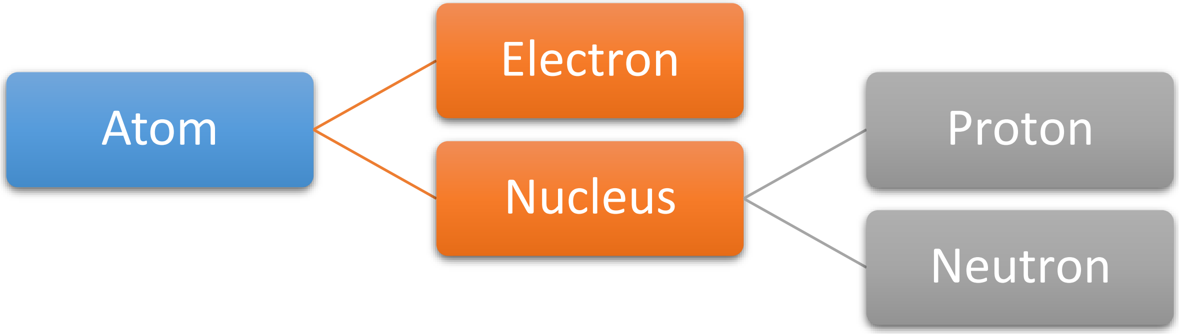
Summary

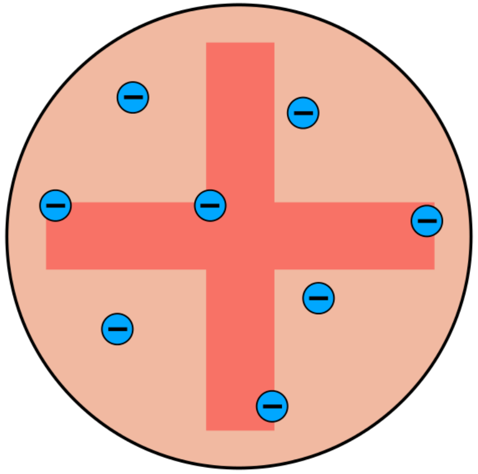
- Stable orbits / states of H atom
- Transitions between the orbits
- Quantized orbital angular momentum
- “Old quantum theory”

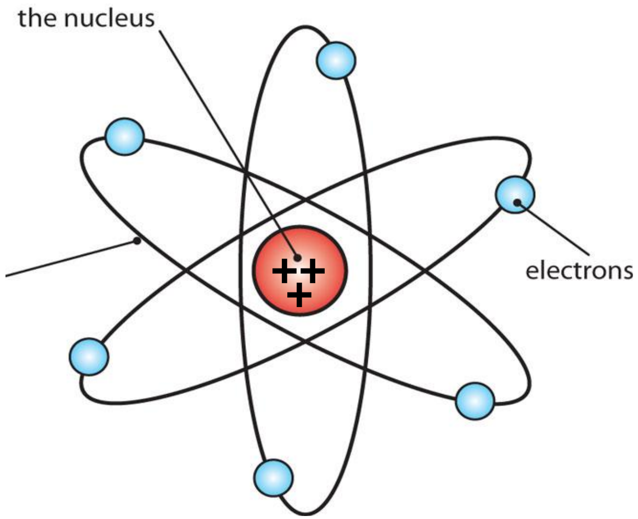


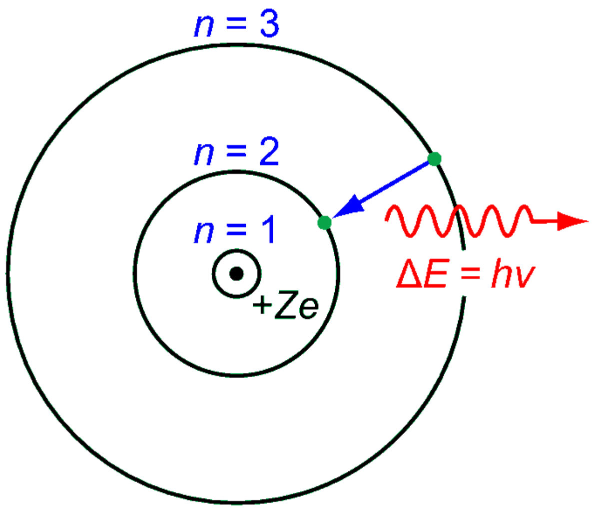
But the Bohr model doesn’t work for the He atom at all!

Summary


Atom
Electron
Nucleus
Proton
Neutron




the nucleus
electrons


n = 3
n = 2
n = 1
+Ze
ΔE = hv

Major points of today's lecture

- **Introduction to quantum mechanics**

1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. The Bohr model: predicting discrete energy levels in atoms
4. Evidence for wave-particle duality
5. The Schrödinger equation
6. Quantum mechanics of particle-in-a-box models
7. Wave functions for particles in two- and three-dimensional boxes

Light: Wave or Particle?

Max Planck
(Berlin, 1858–1947)

Niels Bohr
(Copenhagen, Cambridge
1885–1962)

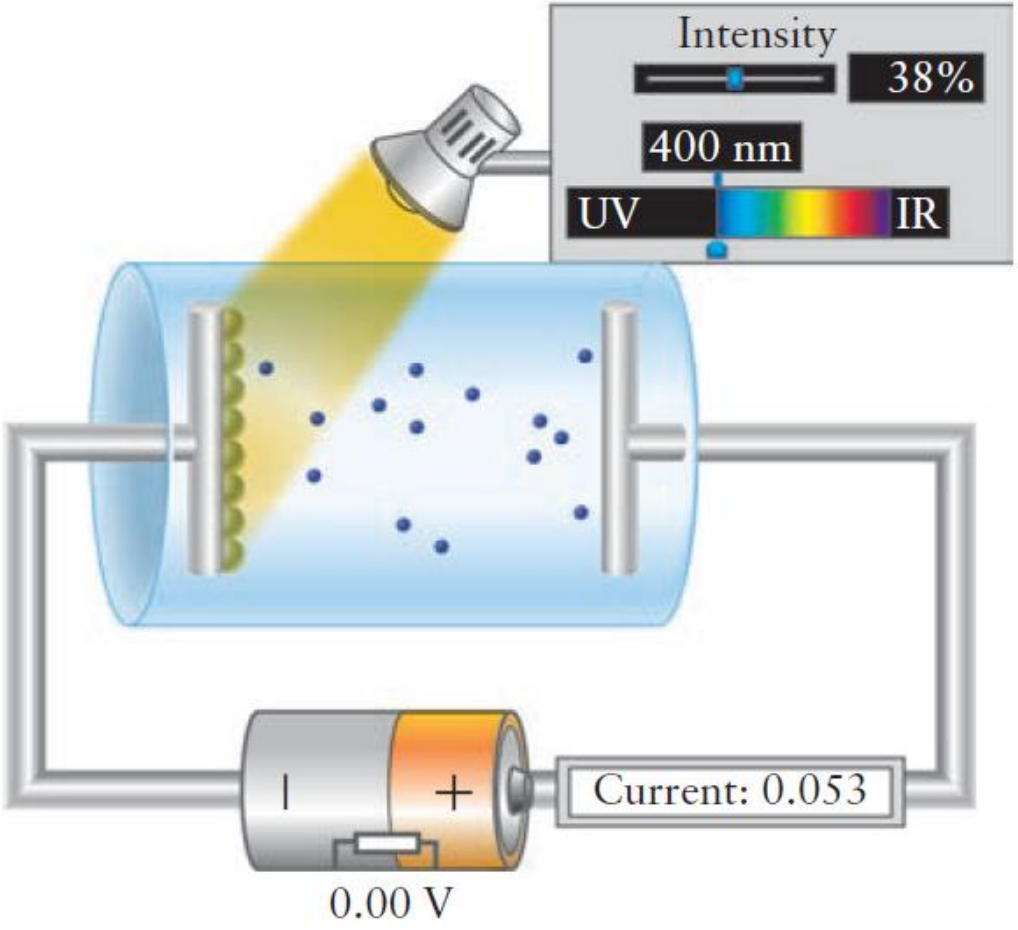
Albert Einstein
(Zürich, 1879–1955)

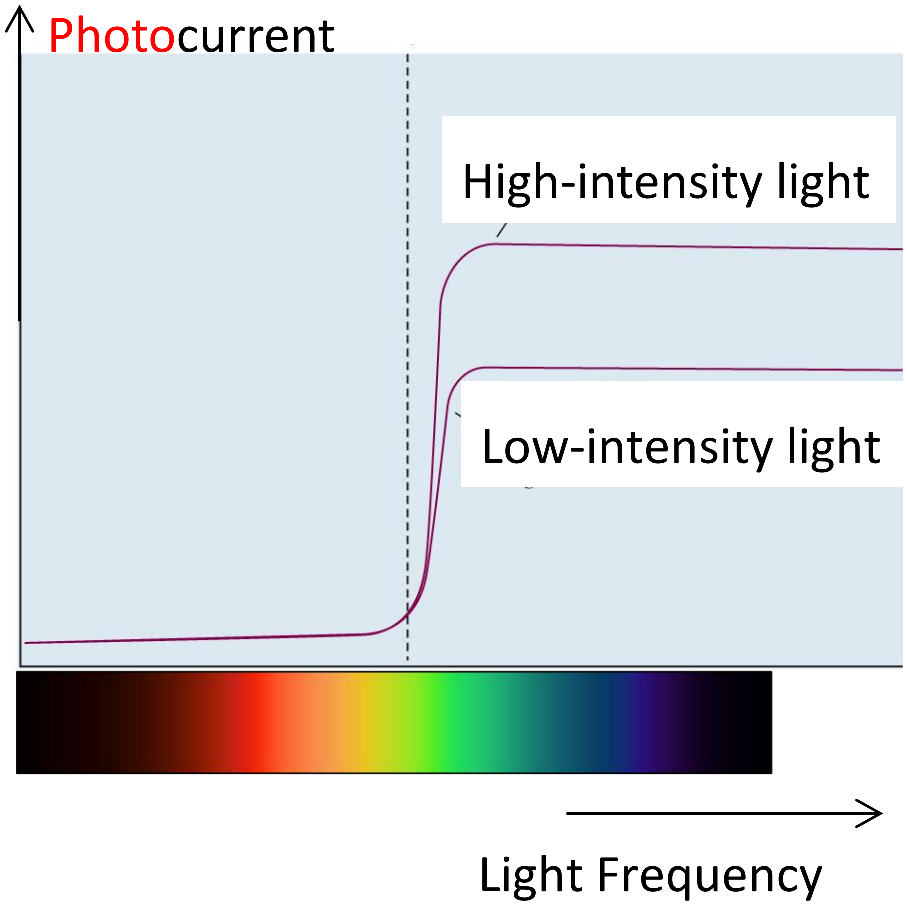
“Energy is quantized in matter but not in light.”

“Light is also quantized.”

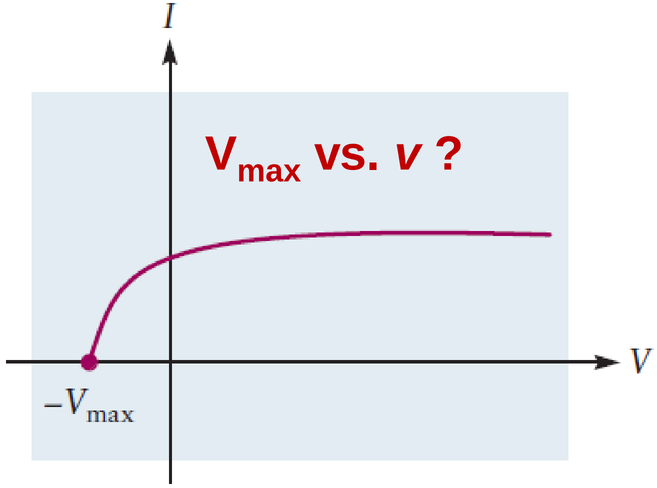
The Photoelectric Effect

The photoelectric effect: a beam of light shinning onto a metal surface (photocathode) can eject electrons (photoelectrons) and cause an electric current (photocurrent) to flow.

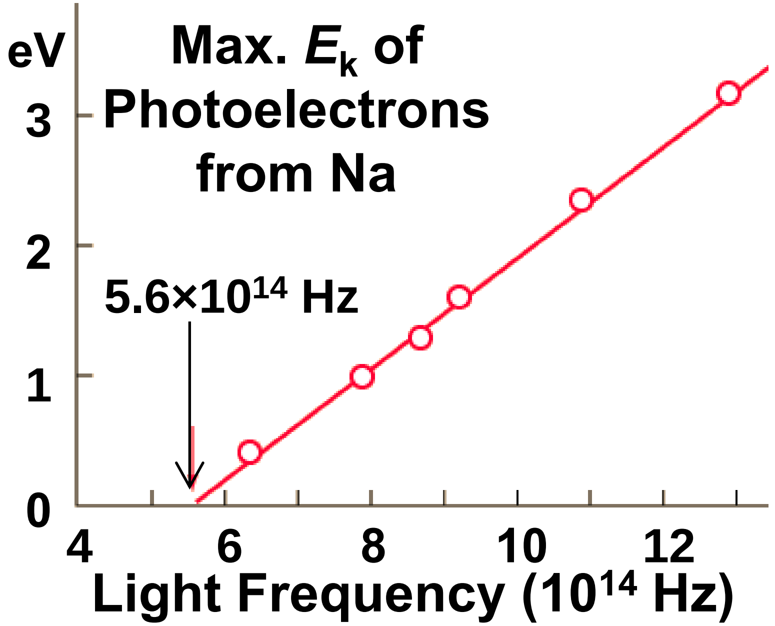




The Photoelectric Effect



The current in a photocell depends on the potential between cathode and collector.



Albert Einstein (1879–1955)

1st: a light wave of frequency *v* consists of quanta of energy (later called photons 光子 by G. N. Lewis), each of which carries energy, $E_{photon} = hv$.

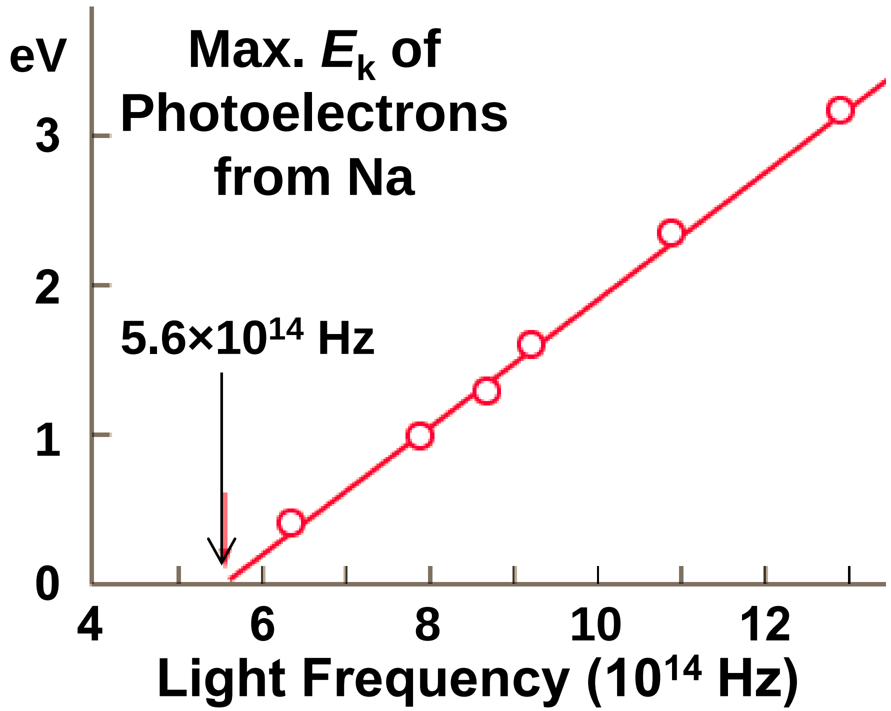
2nd: An electron in the metal absorbs a photon of light and thereby gains the energy required to escape from the metal.

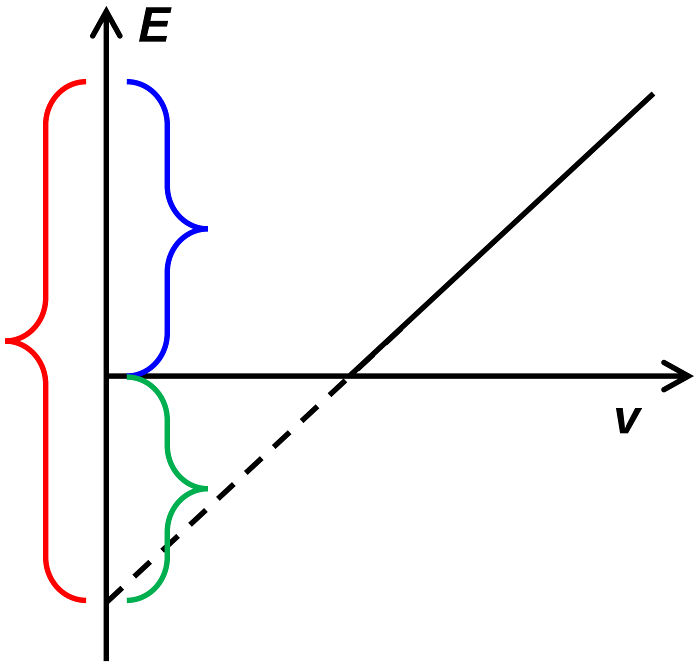
1905: Albert Einstein's Year of Miracles

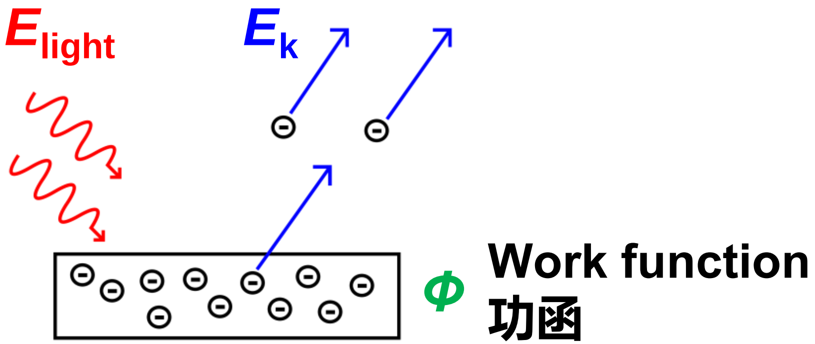
$$E_{light} = hv$$

$$E_{k,max} + \phi = E_{light} \propto v$$

The Photoelectric Effect







1905: Albert Einstein's Year of Miracles

$$E_{k,max} + \Phi = E_{light} \propto v$$

$$E_{light} = hv$$

The Photoelectric Effect

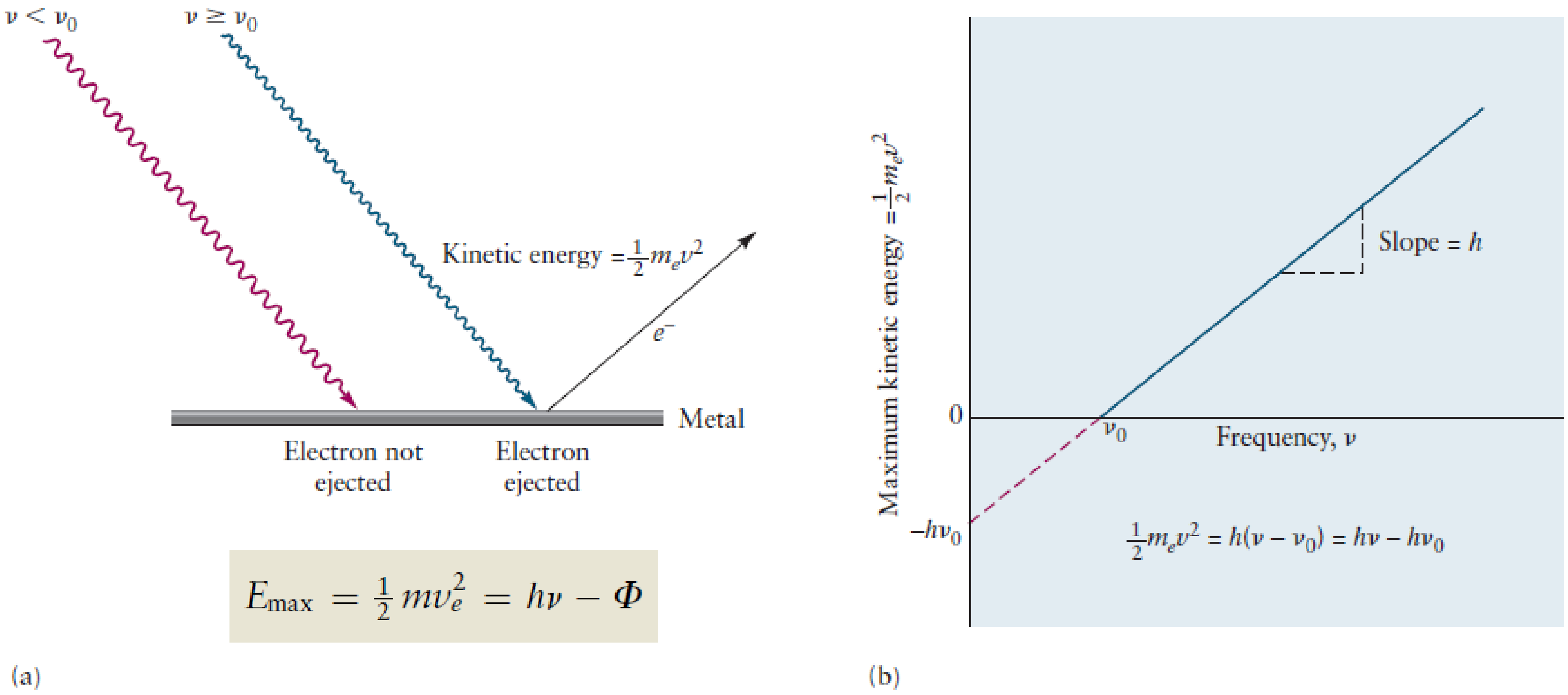


FIGURE 4.18 (a) Two key aspects of the photoelectric effect. Blue light is effective in ejecting electrons from the surface of this metal, but red light is not. (b) The maximum kinetic energy of the ejected electrons varies linearly with the frequency of light used.

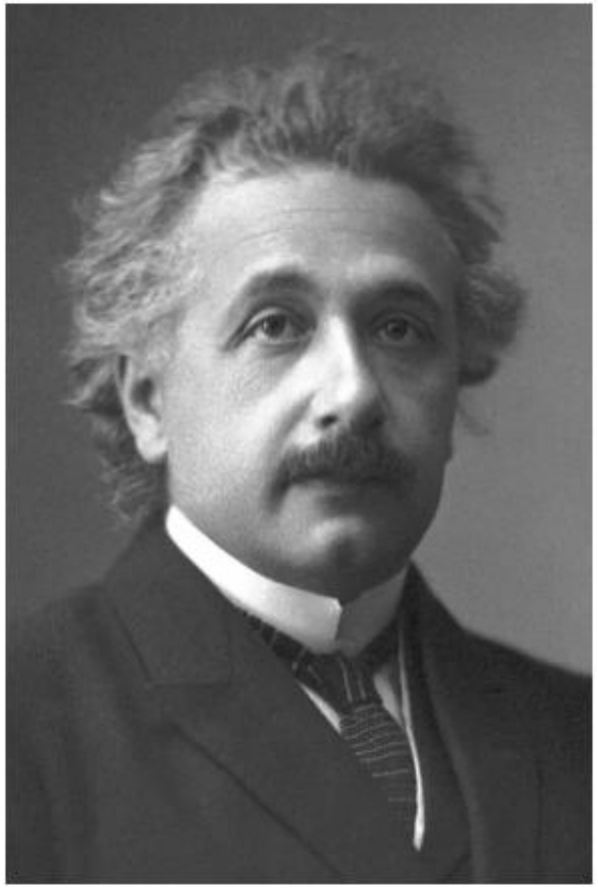


Photo from the Nobel
Foundation archive.

Albert Einstein

Prize share: 1/1

The Nobel Prize in Physics 1921 was awarded to Albert Einstein "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"

https://www.nobelprize.org/prizes/physics/1921/summary/

wave-particle duality of light

(1) Planck 1900, Einstein 1905:

$$E = h\nu$$

(2) Einstein 1916:

$$E = mc^2$$

(3) Definition of momentum

$$p = mv, \quad v = c \text{ for light}$$

(1)+(2)+(3)

$$\boxed{p = \frac{h\nu}{c} = \frac{h}{\lambda}}$$

Photo- + -on = Photon 光子

Question: Light with a wavelength of 400 nm strikes the surface of cesium in a photocell, and the maximum kinetic energy of the electrons ejected is 1.54 × 10^{-19} J. Calculate the work function of cesium and the longest wavelength of light that is capable of ejecting electrons from that metal.

Solution

The frequency of the light is

$$\nu = \frac{c}{\lambda} = \frac{3.00 \times 10^{8} \text{ m s}^{-1}}{4.00 \times 10^{-7} \text{ m}} = 7.50 \times 10^{14} \text{ s}^{-1}$$

The binding energy $h\nu_0$ can be calculated from Einstein's formula:

$$E_{\text{max}} = h\nu - h\nu_0$$

$$1.54 \times 10^{-19} \text{ J} = (6.626 \times 10^{-34} \text{ J s})(7.50 \times 10^{14} \text{ s}^{-1}) - h\nu_0$$

$$= 4.97 \times 10^{-19} \text{ J} - h\nu_0$$

$$\Phi = h\nu_0 = (4.97 - 1.54) \times 10^{-19} \text{ J} = 3.43 \times 10^{-19} \text{ J}$$

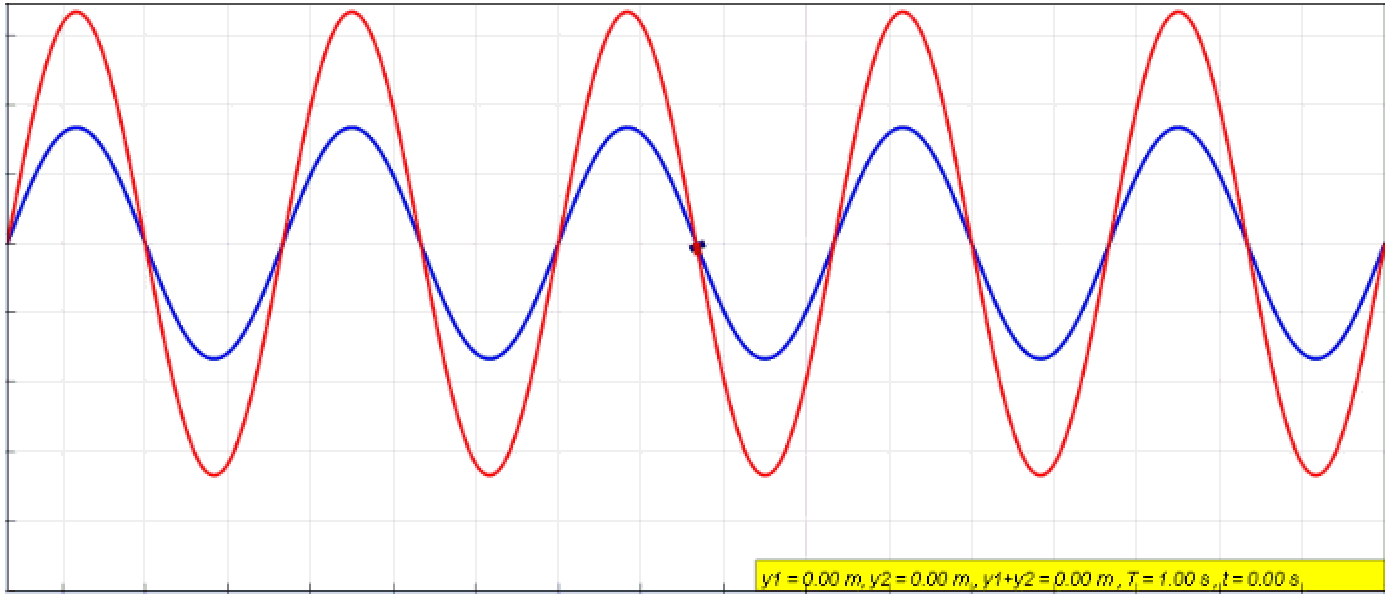
The minimum frequency ν_0 for the light to eject electrons is then

$$\nu_0 = \frac{3.43 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34}} = 5.18 \times 10^{14} \text{ s}^{-1}$$

From this, the maximum wavelength λ_0 is

$$\lambda_0 = \frac{c}{\nu_0} = \frac{3.00 \times 10^{8} \text{ m s}^{-1}}{5.18 \times 10^{14} \text{ s}^{-1}} = 5.79 \times 10^{-7} \text{ m} = 579 \text{ nm}$$

Standing Waves 驻波



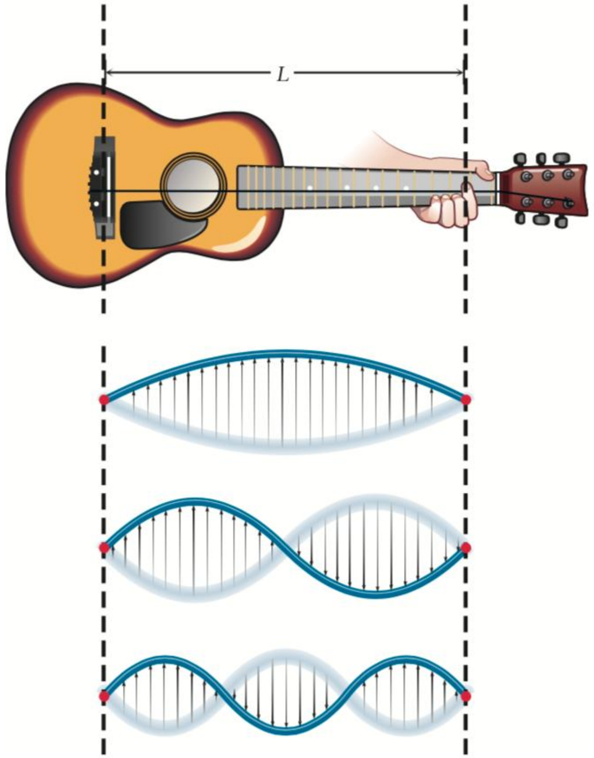


FIGURE 4.17 A guitar string of length L with fixed ends can vibrate in only a restricted set of ways. The positions of largest amplitude for the first three harmonics are shown here. In standing waves such as these, the whole string is in motion except at its end and at the nodes.

- **Equals the sum of two traveling waves in opposite directions;**
- **Does not propagate in either direction and fits in a confined space;**

$$y(x,t) = \frac{A_0}{2}\sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right] + \frac{A_0}{2}\sin\left[2\pi\left(\frac{x}{\lambda} + \frac{t}{T}\right)\right]$$

$$\boxed{y(x,t) = A_0 \sin\left(2\pi\frac{x}{\lambda}\right)\cos\left(2\pi\frac{t}{T}\right).}$$

Standing Waves

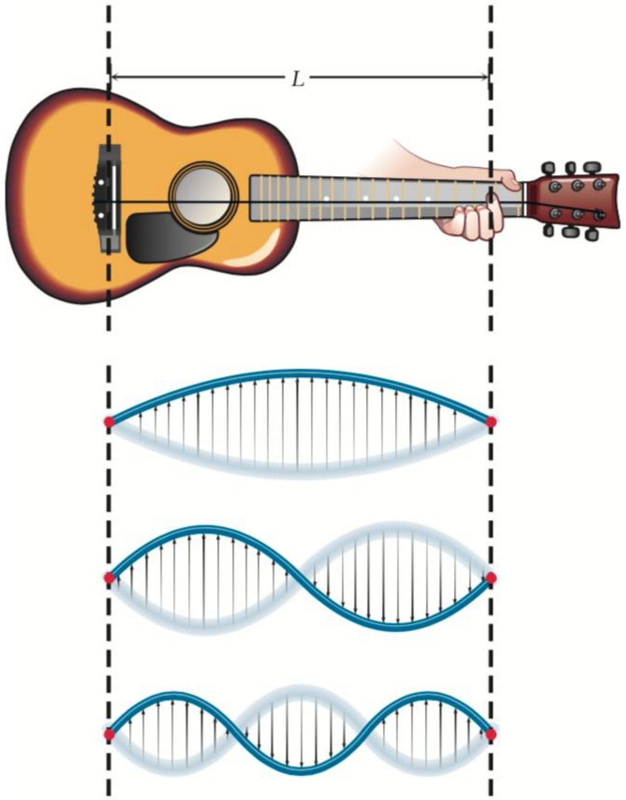


FIGURE 4.17 A guitar string of length L with fixed ends can vibrate in only a restricted set of ways. The positions of largest amplitude for the first three harmonics are shown here. In standing waves such as these, the whole string is in motion except at its end and at the nodes.

The only allowed condition is: an integral number of half-wave-lengths fits into the length of string, L.

$$n\frac{\lambda}{2} = L \qquad n = 1, 2, 3, \ldots$$

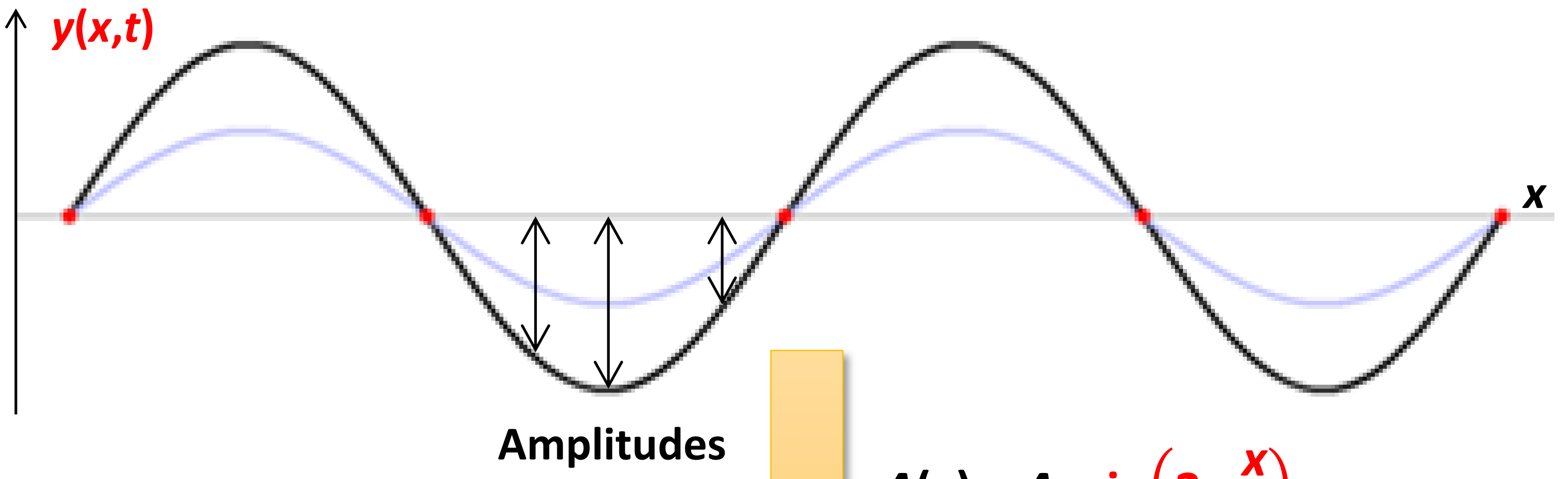
The oscillation with n=1 is called the fundamental or first harmonic （基谐波、一次谐波）, and higher values of n correspond to higher harmonics

Standing Waves

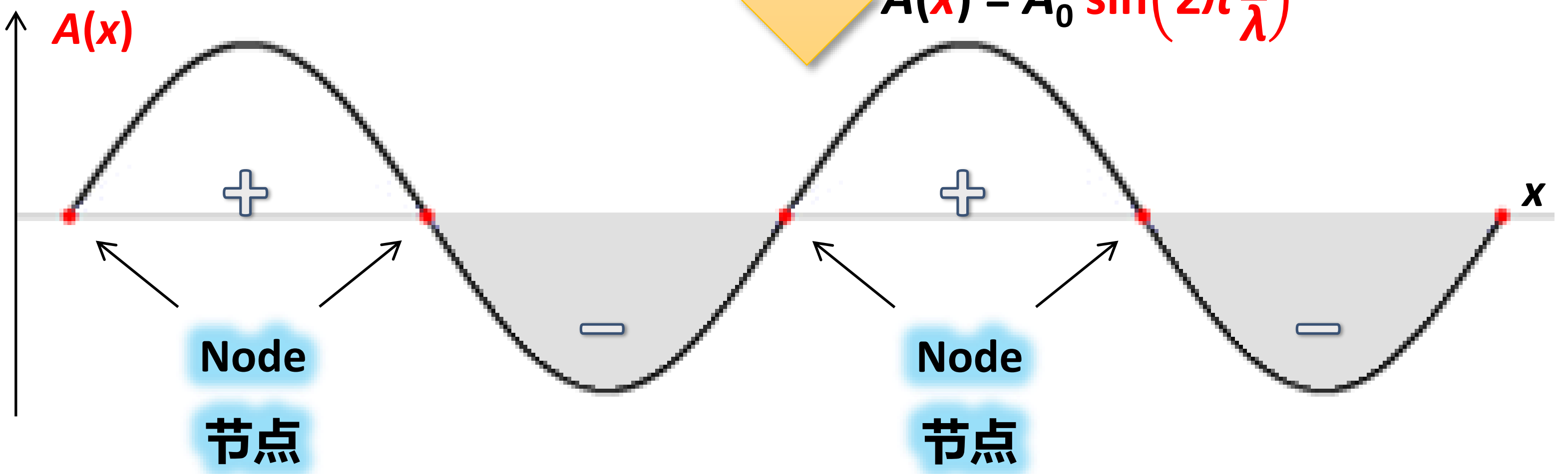
- Changes its amplitude as position varies
- A position with a zero amplitude is called a node

$$y(x,t) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right) \cos\left(2\pi \frac{t}{T}\right)$$

波函数



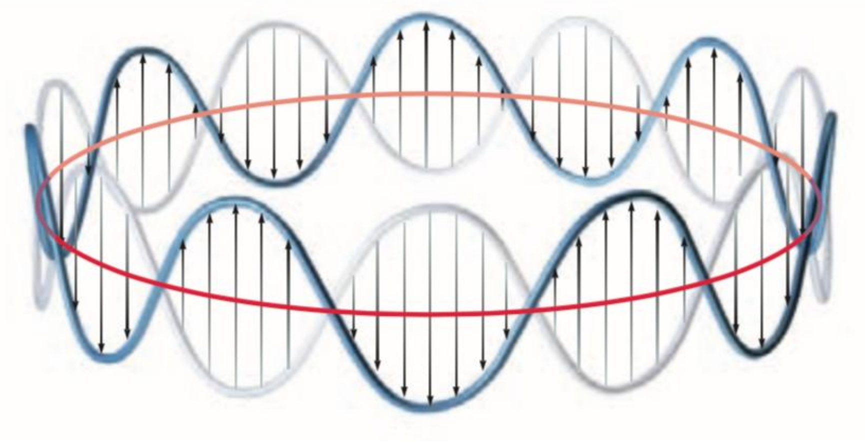
$$A(x) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right)$$



$$y(x,t) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right)$$

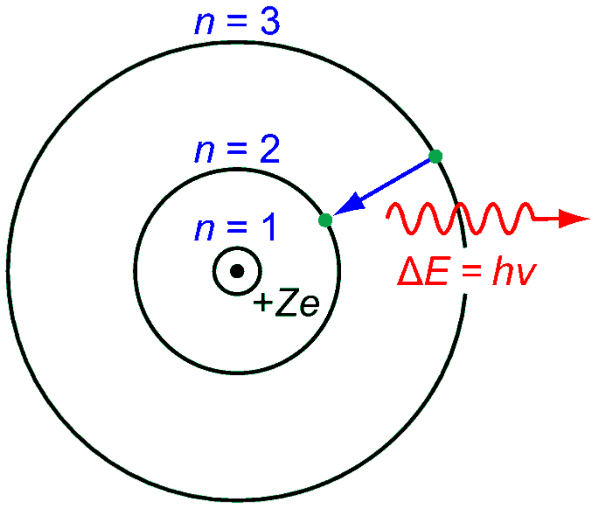
Stationary State 定态

De Broglie Waves



De Broglie realized: **standing waves are examples of quantization**. (only certain discrete vibrational modes, characterized by the "quantum number" n, are allowed for the vibrating string.)

$$n\frac{\lambda}{2} = L \qquad n = 1, 2, 3, \ldots$$

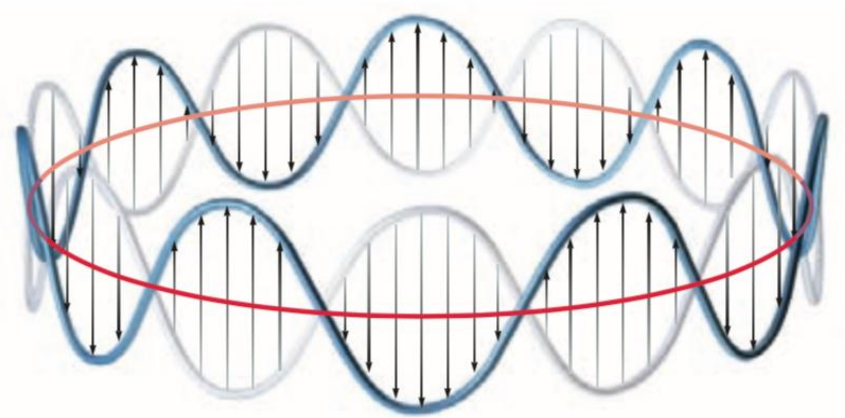


De Broglie suggested: the quantization of energy in a one-electron atom have the same origin. (**the electron might be associated with a circular standing wave oscillating about the nucleus of the atom**.)

$$n\lambda = 2\pi r \qquad n = 1, 2, 3, \ldots$$

Louis de Broglie
(1892–1987)

The wavelength of electron

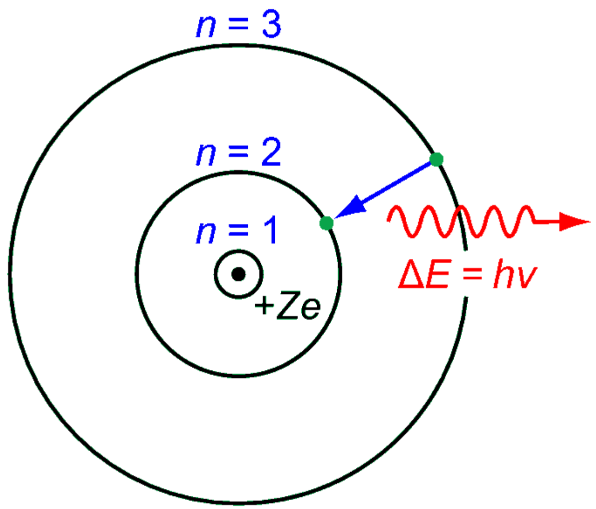


(1) Bohr 1913: angular momentum

$$L = m_e v r = n\frac{h}{2\pi}$$

(2) de Broglie 1924:

$$2\pi r = n\lambda$$

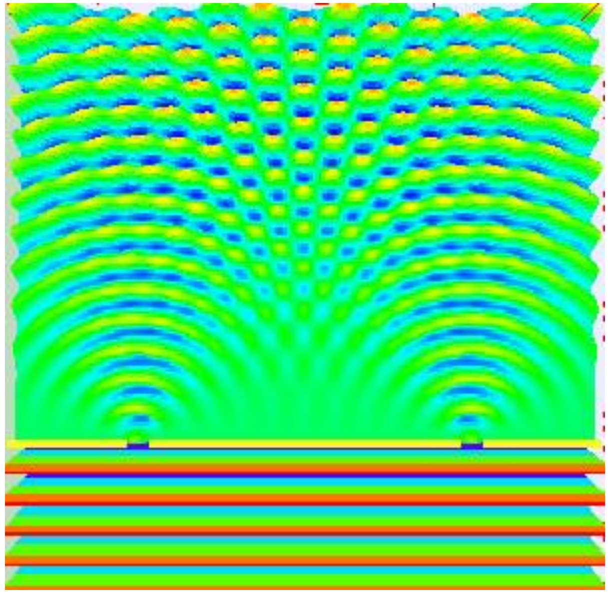


$$\lambda = \frac{h}{m_e v} = \frac{h}{p}$$

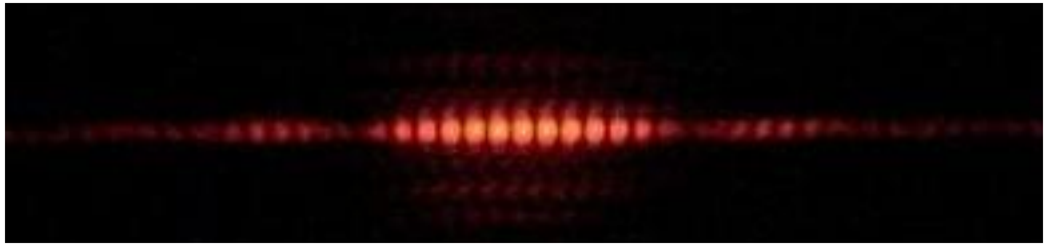
relationship holds for both electron and photon

De Broglie proposed that any particle moving with magnitude of linear momentum p has wavelike properties and a wavelength of h/p associated with its motion.

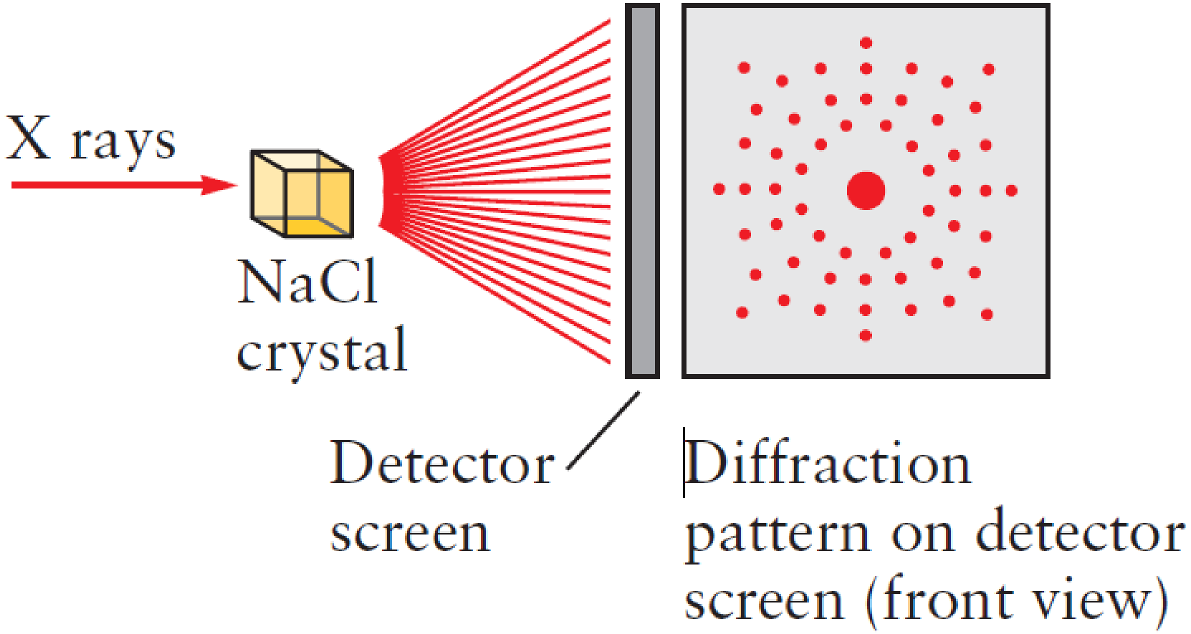
wave-particle duality of electron



Double-slit interference

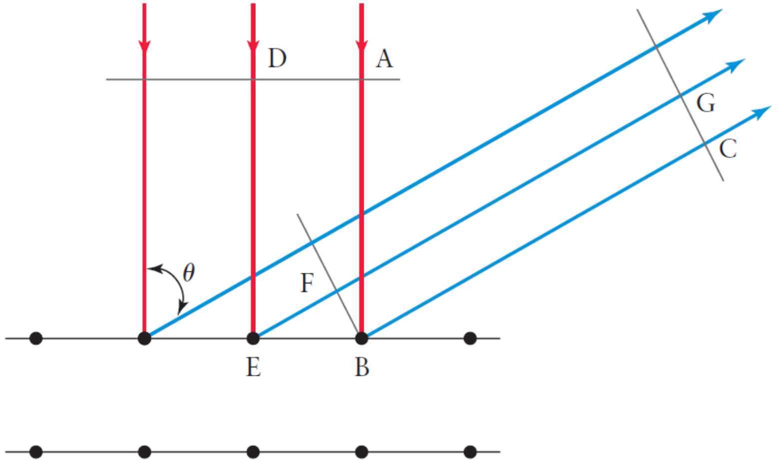


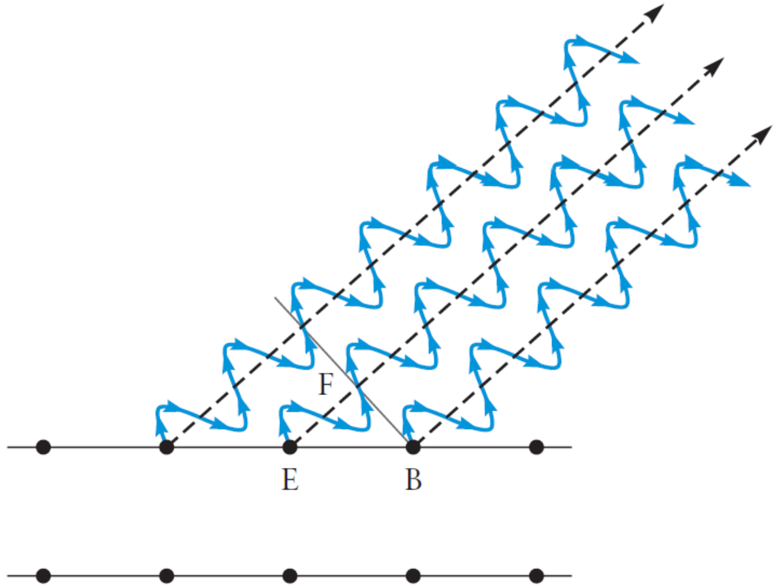
Interference of light

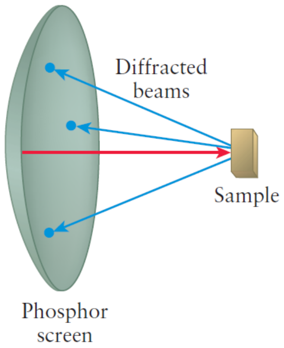


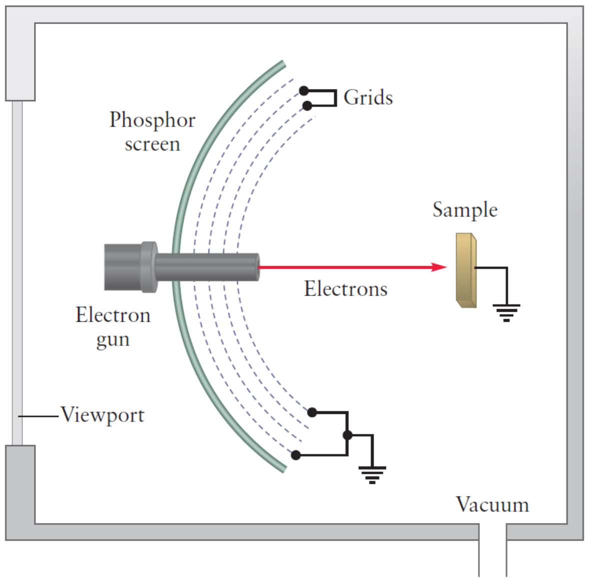
X-ray diffraction by a crystal produces a Laue diffraction pattern.

Electron diffraction

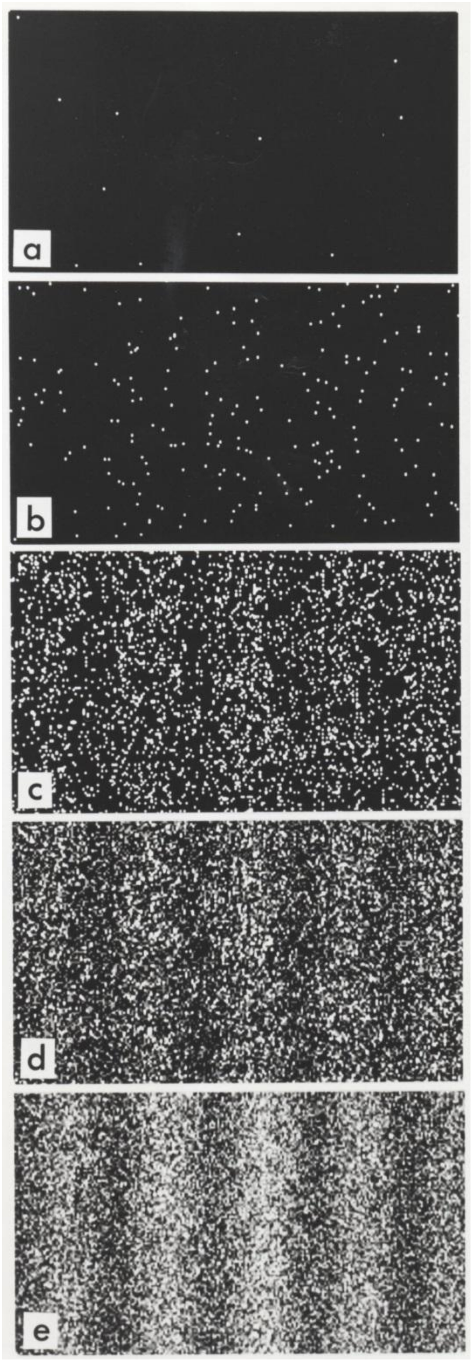








Interference of electrons



$$n\lambda = a \sin \theta$$

Photo from the Nobel Foundation archive.

Prince Louis-Victor Pierre Raymond de Broglie

Prize share: 1/1

The Nobel Prize in Physics 1929 was awarded to Prince Louis-Victor Pierre Raymond de Broglie "for his discovery of the wave nature of electrons"

https://www.nobelprize.org/prizes/physics/1929/summary/

Calculate the de Broglie wavelengths of (a) an electron moving with velocity 1.0×10^6 m s^{-1} and (b) a baseball of mass 0.145 kg, thrown with a velocity of 30 m s^{-1}.

Solution

(a) $$\lambda = \frac{h}{p} = \frac{h}{m_e v} = \frac{6.626 \times 10^{-34}\ \text{J s}}{(9.11 \times 10^{-31}\ \text{kg})(1.0 \times 10^{6}\ \text{m s}^{-1})}$$

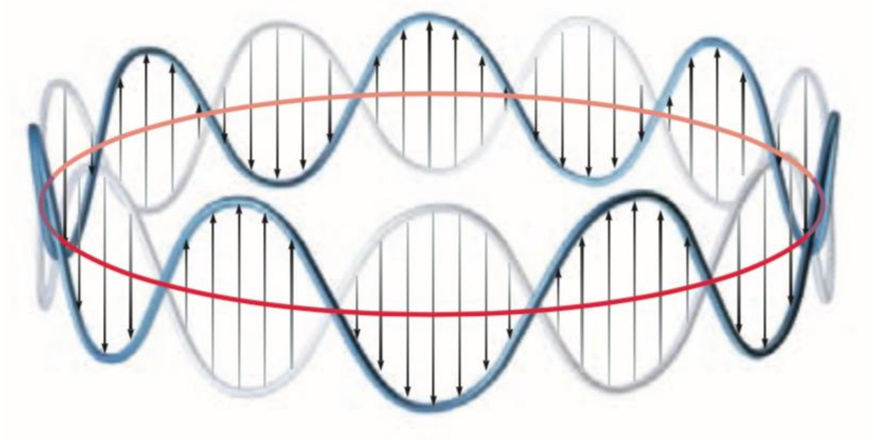
$$= 7.3 \times 10^{-10}\ \text{m} = 7.3\ \text{Å}$$

(b) $$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J s}}{(0.145\ \text{kg})(30\ \text{m s}^{-1})}$$

$$= 1.5 \times 10^{-34}\ \text{m} = 1.5 \times 10^{-24}\ \text{Å}$$

The latter wavelength is far too small to be observed. For this reason, we do not recognize the wavelike properties of baseballs or other macroscopic objects, even though they are always present. However, on a microscopic level, electrons moving in atoms show wavelike properties that are essential for explaining atomic structure.

Indeterminacy and Uncertainty: The Heisenberg Principle



The angular position is indeterminate because it has no definite value: the wave is spread uniformly around the circle, we cannot specify the angular position of the electron on the circle at all.

The Heisenberg indeterminacy principle: when we measure two properties, A and B, the product of which has dimensions of action, we will obtain a spread in results of each identified by ΔA and ΔB that will satisfy the following condition:

$$(\Delta A)(\Delta B) \geq h/4\pi$$

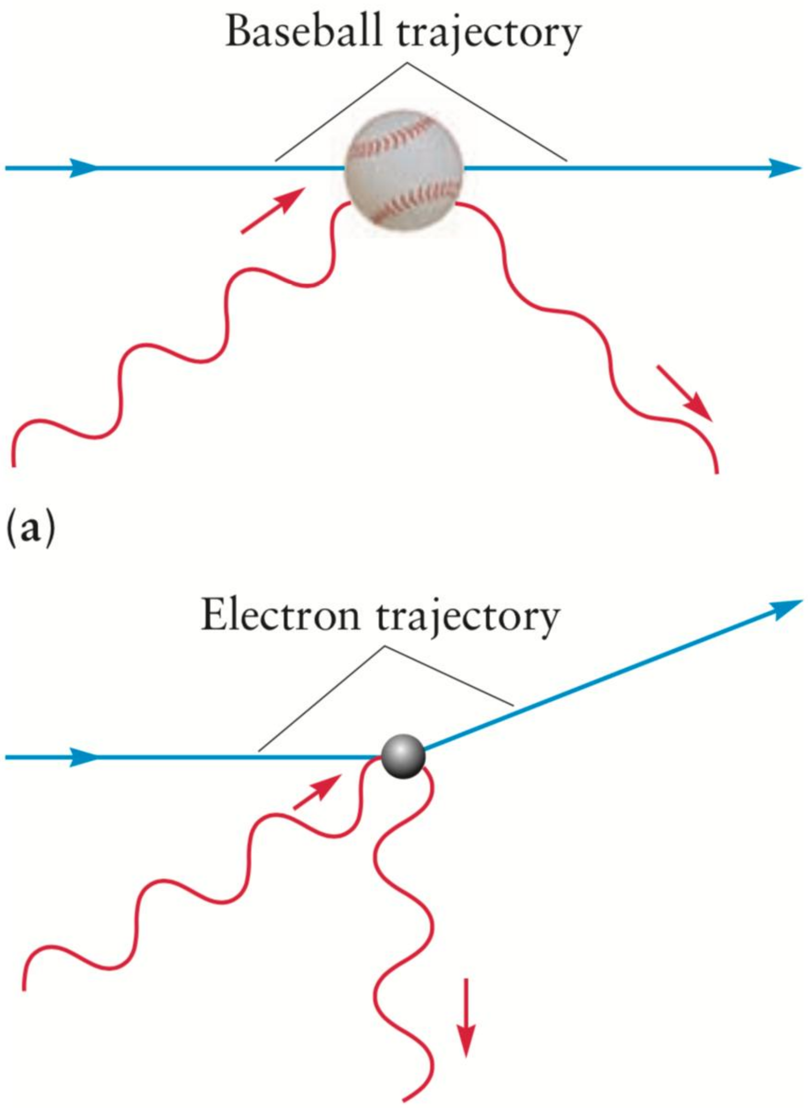
Length (position) × momentum

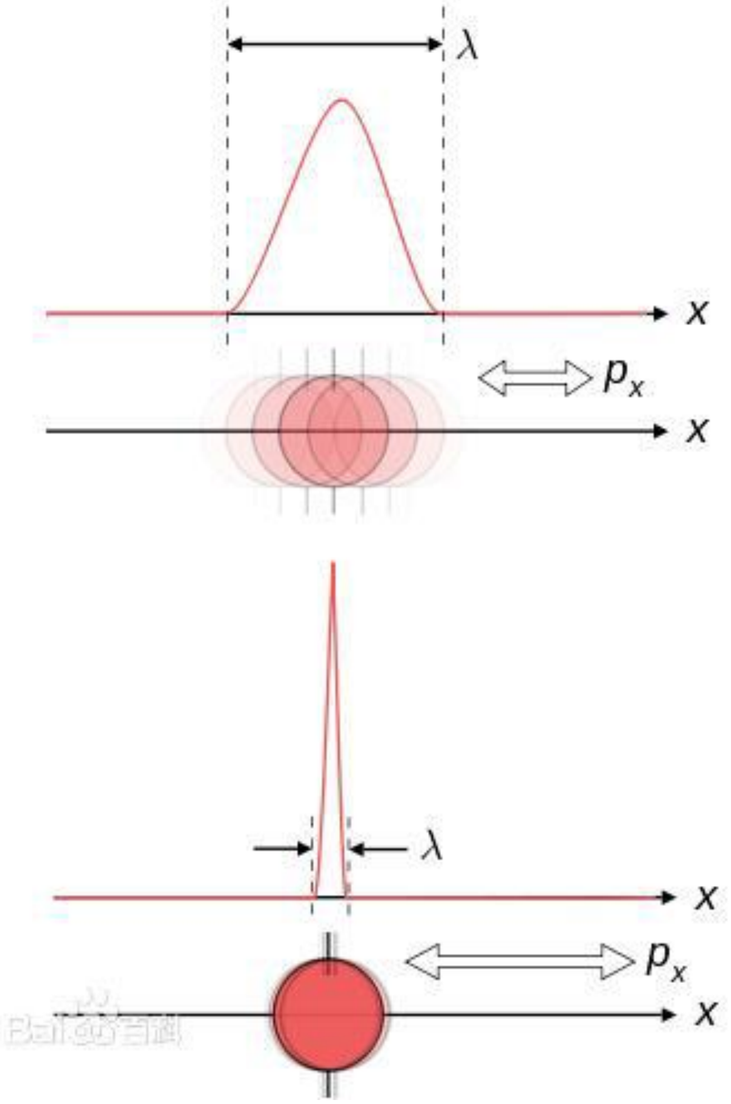
energy × time

Werner Heisenberg

(1901-1976)

Indeterminacy vs. Uncertainty: The Heisenberg Principle





Heisenbeg 1927:

$$\Delta p \cdot \Delta x \geq \frac{h}{4\pi}$$

Standard deviation
标准差

Indeterminacy: observer effect !

Uncertainty: quantum effect !

Photo from the Nobel Foundation archive.

Werner Karl Heisenberg

Prize share: 1/1

The Nobel Prize in Physics 1932 was awarded to Werner Karl Heisenberg "for the creation of quantum mechanics, the application of which has, inter alia, led to the discovery of the allotropic forms of hydrogen"

https://www.nobelprize.org/prizes/physics/1932/summary/

Suppose photons of green light (wavelength 5.3×10^{-7} m) are used to locate the position of the baseball from Example 4.5 with precision of one wavelength. Calculate the minimum spread in the *speed* of the baseball.

Solution

The Heisenberg relation,

$$(\Delta x)(\Delta p) \geq h/4\pi$$

gives

$$\Delta p \geq \frac{h}{4\pi\Delta x} = \frac{6.626 \times 10^{-34} \text{ J s}}{4\pi(5.3 \times 10^{-7} \text{ m})} = 9.9 \times 10^{-29} \text{ kg m s}^{-1}$$

Because the momentum is just the mass (a constant) times the speed, the spread in the speed is

$$\Delta v = \frac{\Delta p}{m} \geq \frac{9.9 \times 10^{-29} \text{ kg m s}^{-1}}{(0.145 \text{ kg})} = 6.8 \times 10^{-28} \text{ m s}^{-1}$$

This is such a tiny fraction of the speed of the baseball (30 m s^{-1}) that indeterminacy plays a negligible role in the measurement of the baseball's motion. Such is the case for all macroscopic objects.

Major points of today's lecture

- **Introduction to quantum mechanics**

1. Preliminaries: wave motion and light
2. Evidence for energy quantization in atoms
3. The Bohr model: predicting discrete energy levels in atoms
4. Evidence for wave-particle duality
5. The Schrödinger equation
6. Quantum mechanics of particle-in-a-box models
7. Wave functions for particles in two- and three-dimensional boxes

The Schrödinger equation

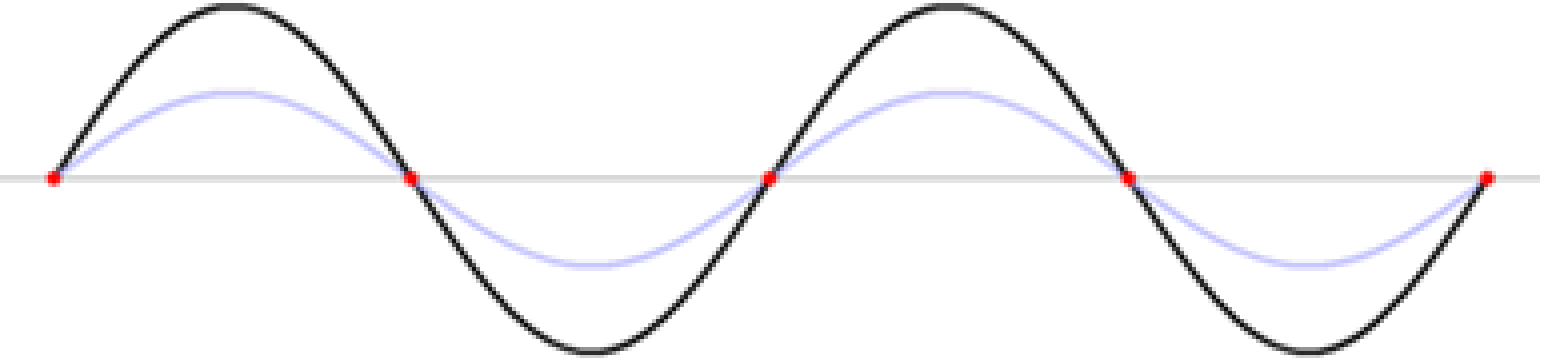
de Broglie 1924: wavelike properties to electron in atom

Schrödinger 1926: electron can be described by a wave function

wave function, psi (ψ): the amplitude of the wave associate with the motion of a particle at (x, y, z), t

Erwin Schrödinger

(1887–1961)



$$y(x,t) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right) \cos\left(2\pi \frac{t}{T}\right).$$

The Schrödinger equation

Find a wave equation that describes the properties of matter on the atomic scale

In 1926, Schrödinger proposed an equation to for finding the wave function of any system: the Schrödinger equation.

$$\hat{H}\Psi = i\hbar\frac{\partial}{\partial t}\Psi$$

- Hamiltonian (哈密顿算符) $\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t).$

$$\sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$$

- Reduced Planck's constant (h-cross or h-bar 约化普朗克常数) $\hbar = h/(2\pi)$

Solve wave function: ψ

The Schrödinger equation

the Schrödinger equation

$$\hat{H}\Psi = i\hbar\frac{\partial}{\partial t}\Psi$$

the Schrödinger equation (time-independent)

$$\hat{H}\psi = E\psi$$

$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t).$$

$$\sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$$

the Schrödinger equation (time-independent, 1D)

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

The Schrödinger equation

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$

Example: for a particle (classical momentum=p) is associated with a wave of wavelength λ=h/p.

The wave functions could be $\psi(x)=A\sin\frac{2\pi x}{\lambda}$ and $\psi(x)=B\cos\frac{2\pi x}{\lambda}$

first derivative $$\frac{d\psi(x)}{dx}=A\frac{2\pi}{\lambda}\cos\frac{2\pi x}{\lambda}$$

second derivative $$\frac{d^2\psi(x)}{dx^2}=-A\left(\frac{2\pi}{\lambda}\right)^2\sin\frac{2\pi x}{\lambda}$$

$$\frac{d^2\psi(x)}{dx^2}=-\left(\frac{2\pi}{\lambda}\right)^2\psi(x)$$

de Broglie relation $$\frac{d^2\psi(x)}{dx^2}=-\left(\frac{2\pi}{h}p\right)^2\psi(x) \Rightarrow -\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2}=\frac{p^2}{2m}\psi(x)$$

The Schrödinger equation

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} = \frac{p^2}{2m}\psi(x)$$

kinetic energy $\mathcal{T} = p^2/2m$

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} = \frac{p^2}{2m}\psi(x) = \mathcal{T}\psi(x)$$

$E = \mathcal{T} + V(x)$ **kinetic energy + potential energy**

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

time-independent Schrödinger equation for a particle moving in one dimension.

$\psi(x) = A \sin \frac{2\pi x}{\lambda}$ and $\psi(x) = B \cos \frac{2\pi x}{\lambda}$ $\rightleftarrows$ $-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$

Photo from the Nobel Foundation archive.

Erwin Schrödinger

Prize share: 1/2

Photo from the Nobel Foundation archive.

Paul Adrien Maurice Dirac

Prize share: 1/2

The Nobel Prize in Physics 1933 was awarded jointly to Erwin Schrödinger and Paul Adrien Maurice Dirac "<u>for the discovery of new productive forms of atomic theory</u>"

https://www.nobelprize.org/prizes/physics/1933/summary/

Interpretation of the wave function in the Schrödinger equation

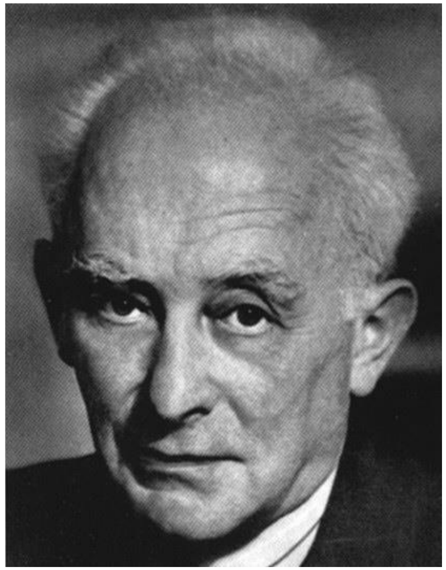
What is the **physical meaning of the wave function ψ?** (no way of measuring ψ directly)

classical theory of electromagnetism (wave view):

- No direct way to measure the amplitudes of the electric and magnetic fields of light wave.
- Intensity of the light wave (can be measured) is proportional to the square of the amplitude of the electric field: $\text{intensity} \propto (E_{\max})^2$

Quantum mechanics (particle view):

- the intensity is proportional to the density of photons in a region of space.



Max Born

(1882-1970)

Connecting the **wave and particle views of the electromagnetic field**: probability of finding a photon is given by the square of the amplitude of the electric field

Probability of finding a particle is given by the square of the amplitude of the wave function

Probability: $P(x)= |\psi(x)|^2$

Interpretation of the wave function in the Schrödinger equation

Probability of finding a particle is given by the square of the amplitude of the wave function

Probability: $P(x) = |\psi(x)|^2$

$[\psi(x, y, z)]^2\, dV$ is the probability that the particle will be found in a small volume $dV = dxdydz$ centered at the point (x, y, z)

Copenhagen interpretation 哥本哈根诠释

Interpretation of the wave function in the Schrödinger equation

The probabilistic interpretation requires that any function must meet three mathematical conditions before it can be used as a wave function.

“单值“

- **The probability density must be normalized, ensures that probability density is properly defined, and that all possible outcomes are included （平方可积）**

$$\int_{-\infty}^{+\infty} P(x)dx = \int_{-\infty}^{+\infty} [\psi(x)]^2 dx = 1$$

- **P(x) must be continuous at each point x.（连续）**

At some specific point, its value must be the same regardless approached from the left or from the right. This translates into the condition that ψ(x) and its first derivative ψ’(x) are continuous.

- **P(x) must be bounded at large values of x.（有限）**

$$\psi \longrightarrow 0 \quad \text{as} \quad x \longrightarrow \pm\infty$$

Photo from the Nobel Foundation archive.

Max Born

Prize share: 1/2

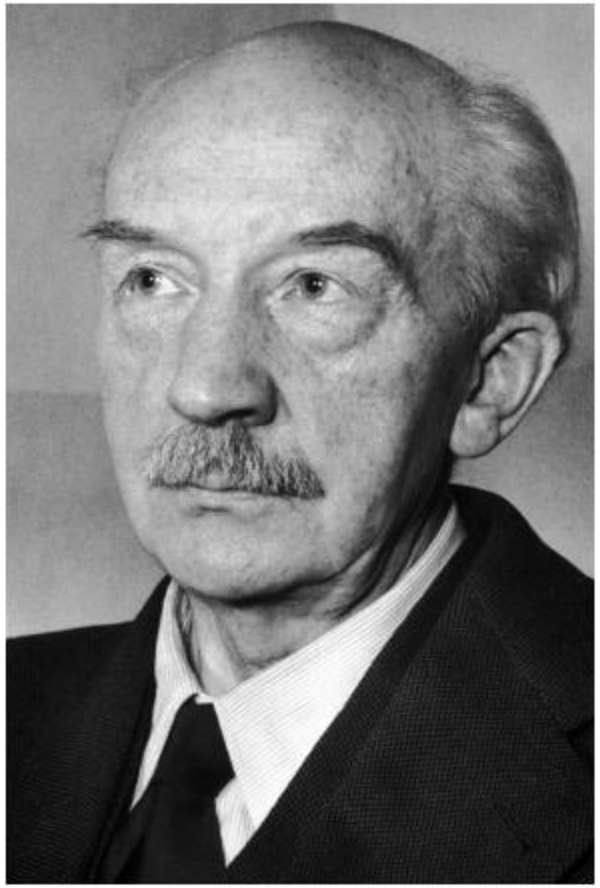


Photo from the Nobel Foundation archive.

Walther Bothe

Prize share: 1/2

The Nobel Prize in Physics 1954 was divided equally between Max Born "for his fundamental research in quantum mechanics, especially for his statistical interpretation of the wavefunction" and Walther Bothe "for the coincidence method and his discoveries made therewith"

https://www.nobelprize.org/prizes/physics/1954/summary/

Major points of today's lecture

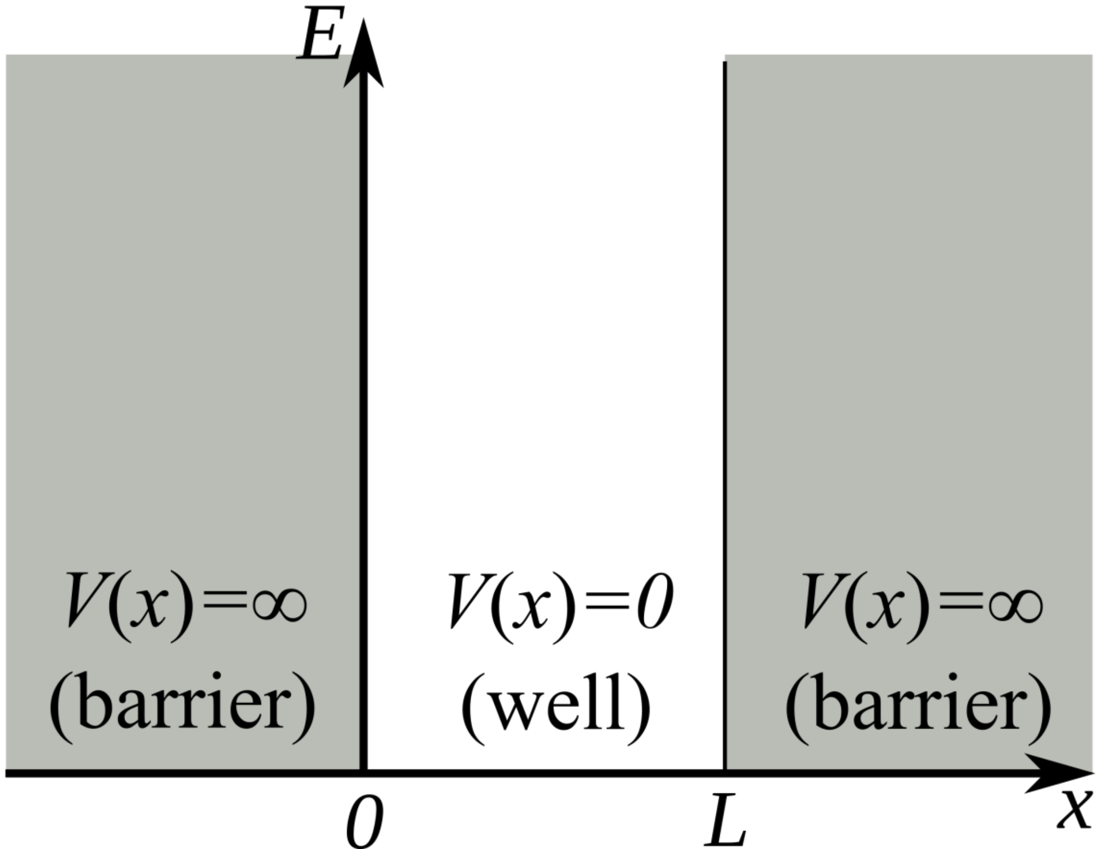
- **Introduction to quantum mechanics**
  1. Preliminaries: wave motion and light
  2. Evidence for energy quantization in atoms
  3. The Bohr model: predicting discrete energy levels in atoms
  4. Evidence for wave-particle duality
  5. The Schrödinger equation
  6. Quantum mechanics of particle-in-a-box models
  7. Wave functions for particles in two- and three-dimensional boxes

Particle-in-a-Box models

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Particle-in-a-Box models: a particle confined by potential energy barriers to a certain region of space.

1D box: the particle is located on the x-axis in the interval between 0 and L (L is the length of the box)



- The potential energy $V(x)$ rise abruptly to an infinite value at the two end walls to prevent fast-moving particles from escaping.
- Inside the box, the motion of the particle is free, so $V(x) = 0$

$$V(x) = \begin{cases} 0, & x_c - \frac{L}{2} < x < x_c + \frac{L}{2}, \\ \infty, & \text{otherwise}, \end{cases},$$

$$\psi(x) = 0 \text{ for } x \leq 0 \text{ and } x \geq L$$

Particle-in-a-Box models

Schrödinger Equation for a particle in a box:

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Inside the box, where $V = 0$, the Schrödinger equation has the following form:

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

or equivalently,

$$\frac{d^2\psi(x)}{dx^2} = -\frac{8\pi^2 mE}{h^2}\psi(x)$$

Particle-in-a-Box models

a potential solution? $$\psi(x) = A \sin kx$$

- **The boundary conditions require: ψ(x) = 0 at x = 0** ➡
  - **The cosine function can be eliminated, cos(0) = 1**
  - **The sine function meets the requirement, sin(0) = 0**

- **The boundary conditions require that ψ(x) = 0 at x = L** ➡

$$\psi(L) = 0$$

$$\psi(L) = A \sin kL = 0$$

$$kL = n\pi \qquad n = 1, 2, 3, \ldots$$

$$\psi(x) = A \sin\left(\frac{n\pi x}{L}\right) \qquad n = 1, 2, 3, \ldots$$

Particle-in-a-Box models

The wave function must be normalized

$$A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = 1$$

$$A^2\left(\frac{L}{2}\right) = 1 \qquad A = \sqrt{\frac{2}{L}}$$

The normalized wave function for the one-dimensional particle in a box is

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) n = 1, 2, 3, \ldots}$$

Particle-in-a-Box models

$$-\frac{h^2}{8\pi^2 m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{8\pi^2 mE}{h^2}\psi(x)$$

$$\begin{aligned}\frac{d^2\psi_n(x)}{dx^2} &= \frac{d^2}{dx^2}\left[\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)\right]\\ &= -\left(\frac{n\pi}{L}\right)^2\left[\sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)\right]\\ &= -\left(\frac{n\pi}{L}\right)^2\psi_n(x)\end{aligned}$$

$$\frac{8\pi^2 mE_n}{h^2} = \frac{n^2\pi^2}{L^2}$$

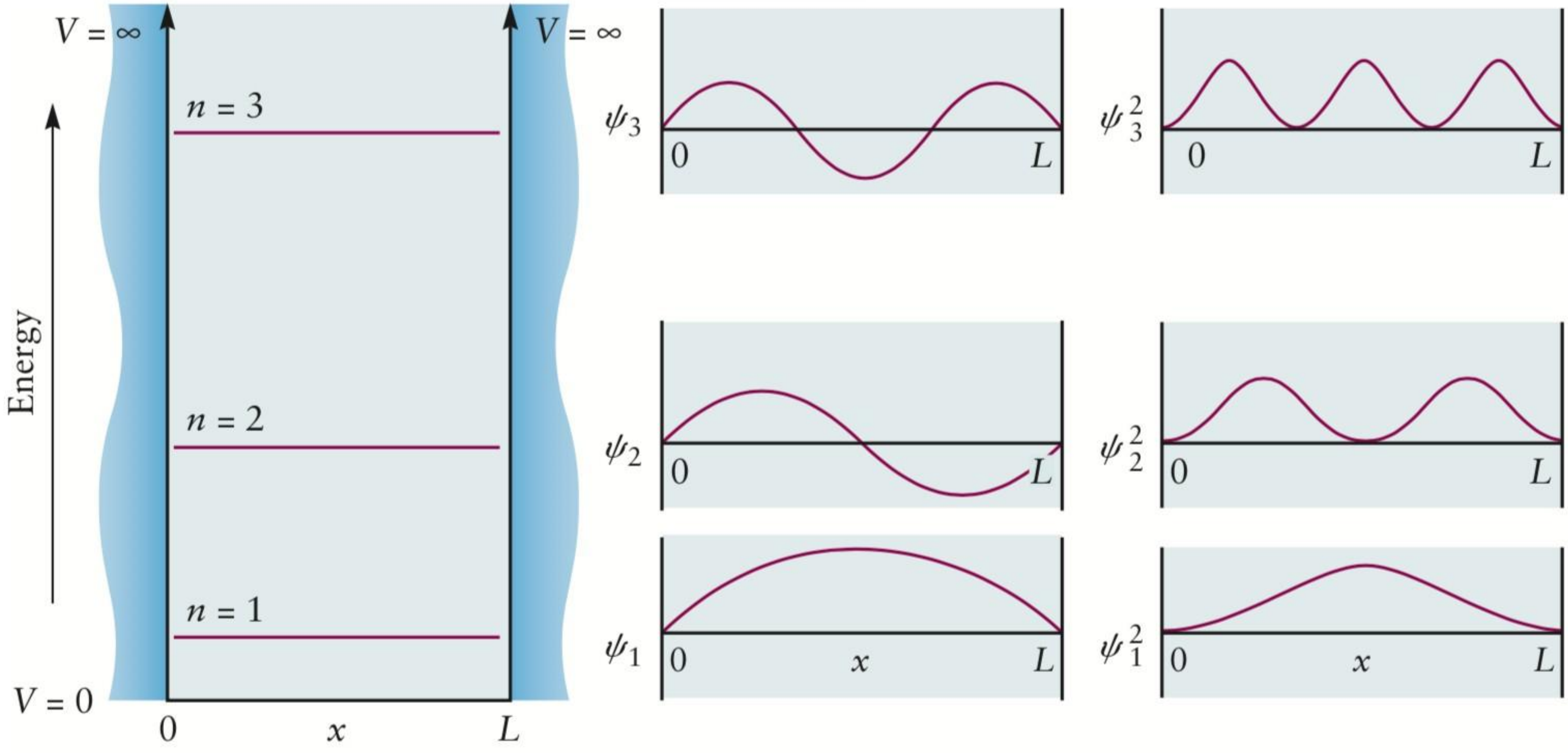
$$E_n = \frac{n^2h^2}{8mL^2} \qquad n = 1, 2, 3, \ldots$$

The energy of a particle in the box is quantized

Particle-in-a-Box models

The normalized wave function for the one-dimensional particle in a box is

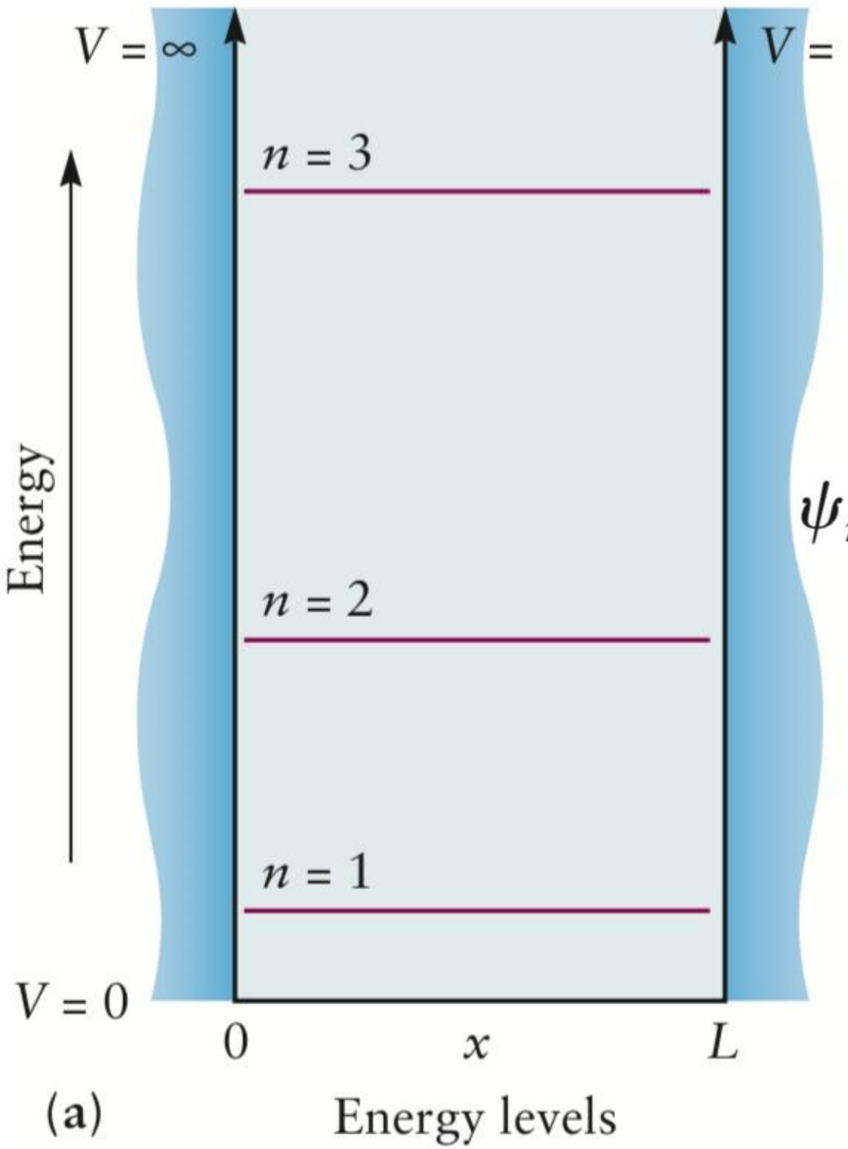
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad n = 1, 2, 3, \ldots$$



(a) Energy levels (b) Wave functions (c) Probabilities

The wave function ψ_n has n - 1 nodes

Particle-in-a-Box models



Can a particle in a box have zero energy?

If n = 0; E_0=0;

$\psi_n(x) = A \sin(n\pi x/L)$ makes $\psi_0(x)$ zero everywhere

no particle in the box

Particle-in-a-Box models

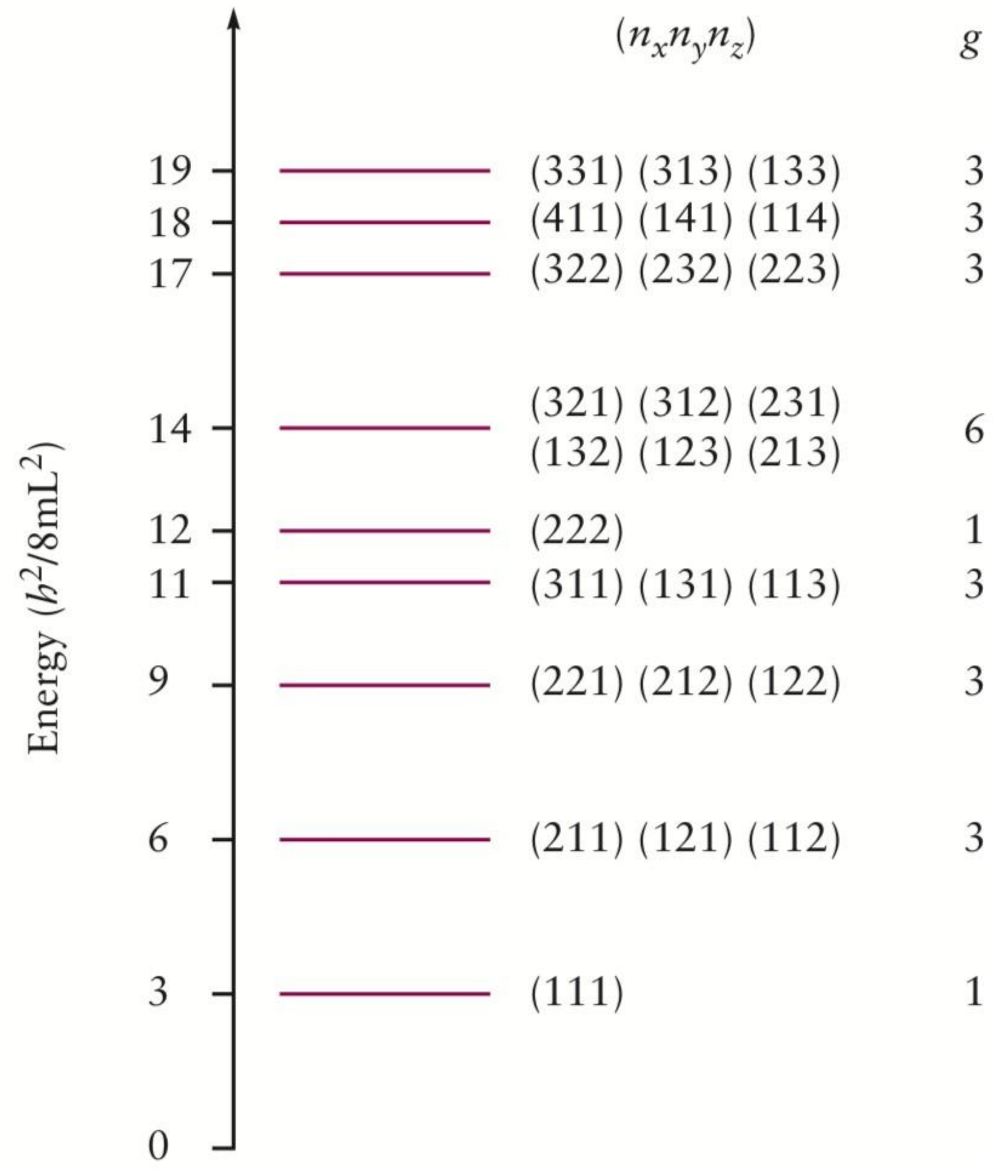
3-D boxes?

$$E_{n_x n_y n_z} = \frac{h^2}{8m}\left[\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}\right] \qquad \begin{cases} n_x = 1, 2, 3, \ldots \\ n_y = 1, 2, 3, \ldots \\ n_z = 1, 2, 3, \ldots \end{cases}$$

Cubic box by setting $L_x = L_y = L_z = L$.

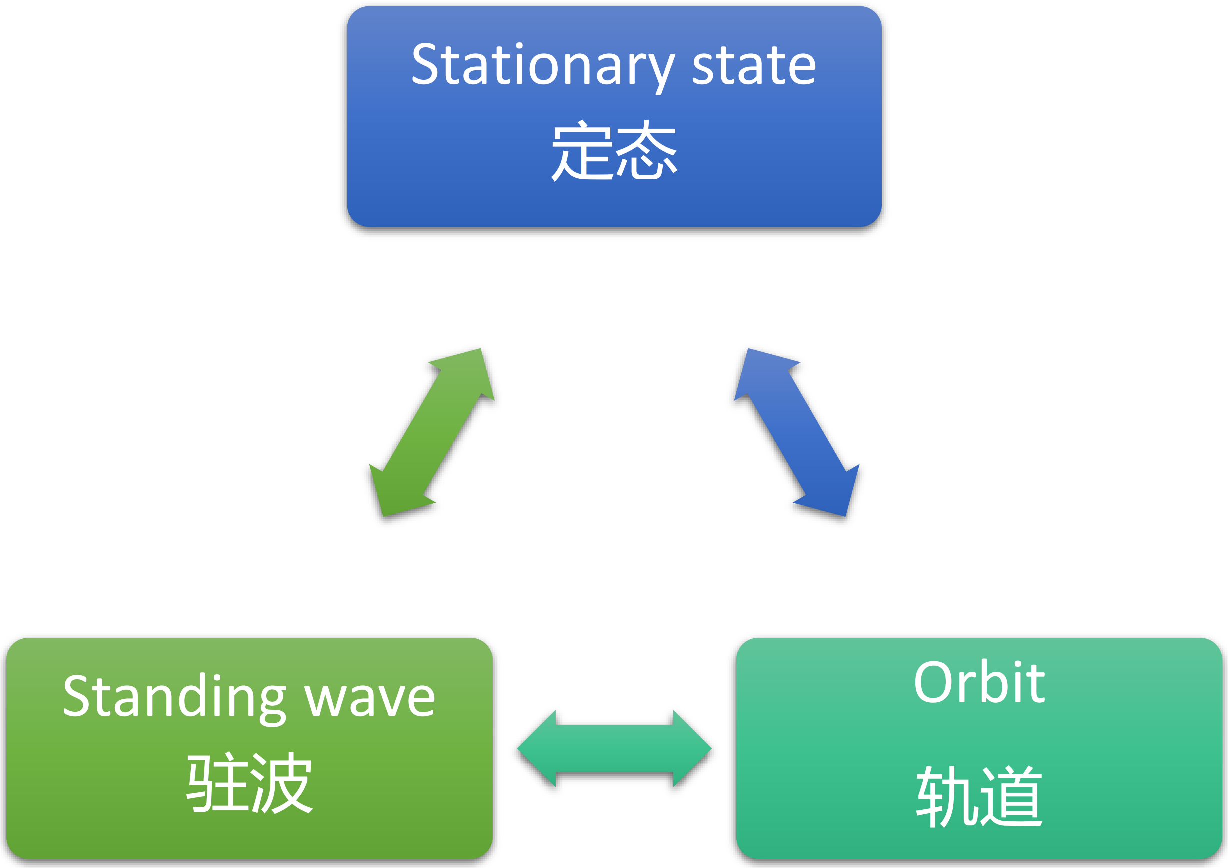
$$E_{n_x n_y n_z} = \frac{h^2}{8mL^2}\left[n_x^2 + n_y^2 + n_z^2\right] \qquad \begin{cases} n_x = 1, 2, 3, \ldots \\ n_y = 1, 2, 3, \ldots \\ n_z = 1, 2, 3, \ldots \end{cases}$$

Particle-in-a-Box models



- **Certain energy values appear more than once (squares of different sets of quantum numbers can add up to give the same total).**
- **Such energy levels, which correspond to more than one quantum state, are called degenerate（简并）.**

Summary



Consider the following two systems: (a) an electron in a one-dimensional box of length 1.0 Å and (b) a helium atom in a cube 30 cm on an edge. Calculate the energy difference between ground state and first excited state, expressing your answer in kJ mol^{-1}.

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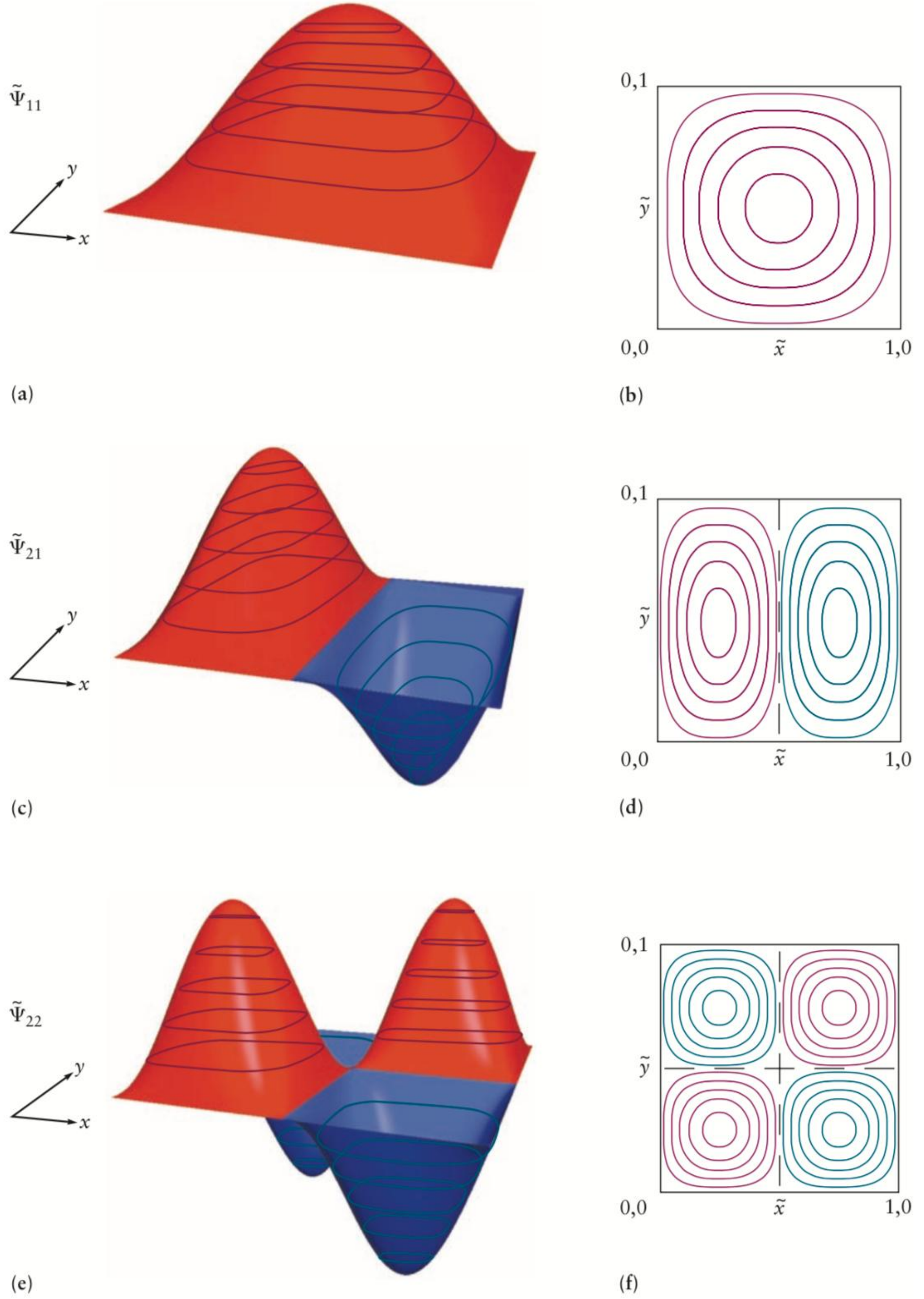
Particle in square boxes

$$\Psi_{n_x n_y}(x,\ y) = \psi_{n_x}(x)\psi_{n_y}(y) = \frac{2}{L}\ \sin\left(\frac{n_x \pi x}{L}\right)\ \sin\left(\frac{n_y \pi y}{L}\right)$$

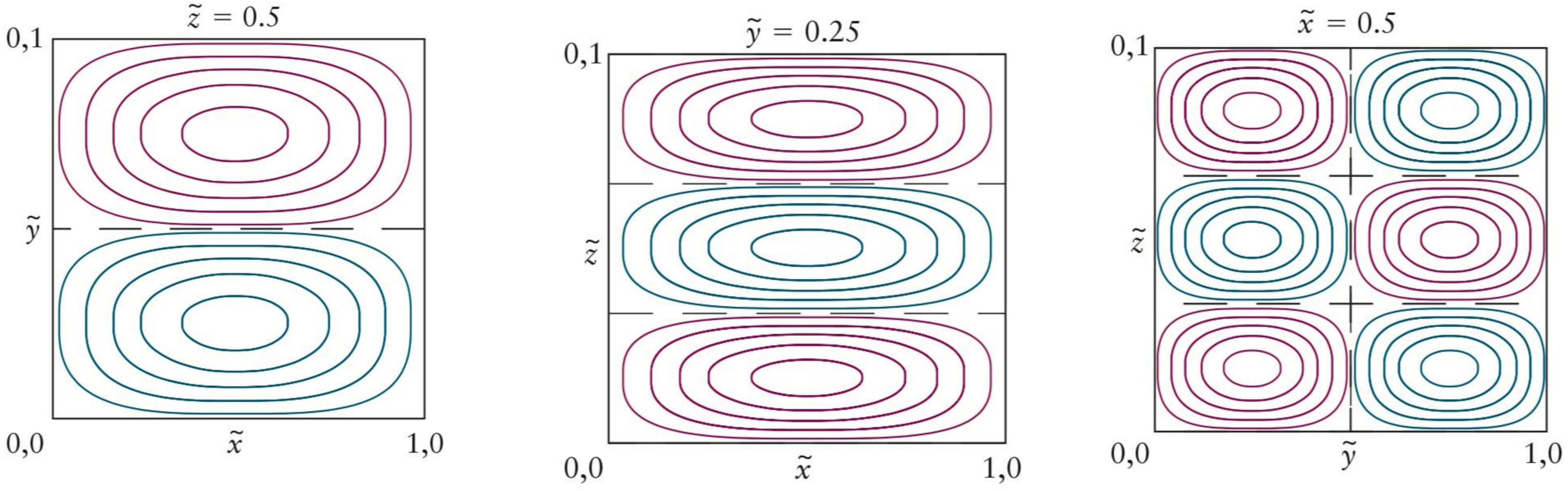
The normalized wave function for the one-dimensional particle in a box is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) n = 1, 2, 3, \ldots$$



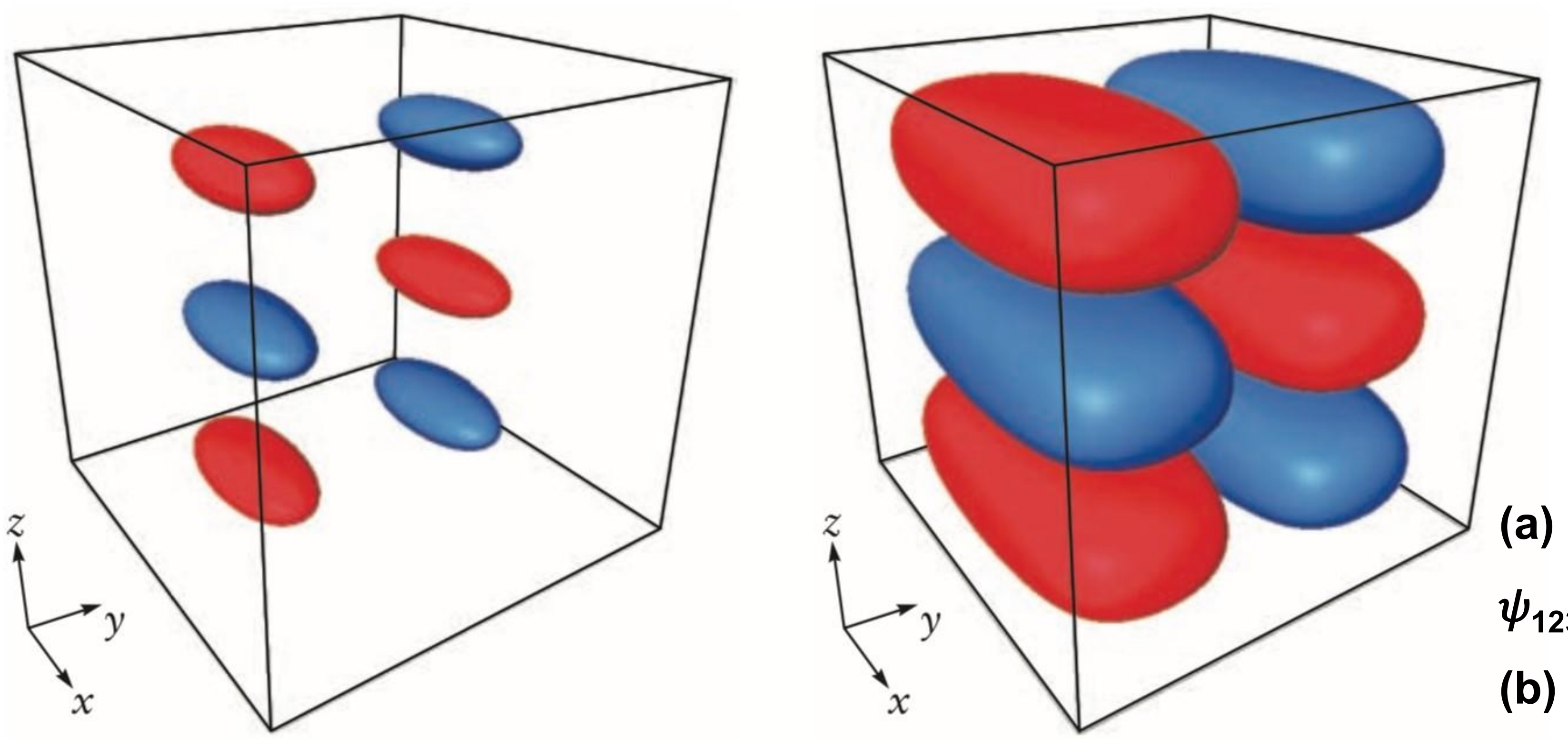
Particle in cubic boxes

$$\Psi_{n_x n_y n_z}(x, y, z) = \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{n_x \pi x}{L}\right) \sin\left(\frac{n_y \pi y}{L}\right) \sin\left(\frac{n_z \pi z}{L}\right)$$



Particle in cubic boxes

Isosurfaces (等值面): the wave function has constant value at each point on them.

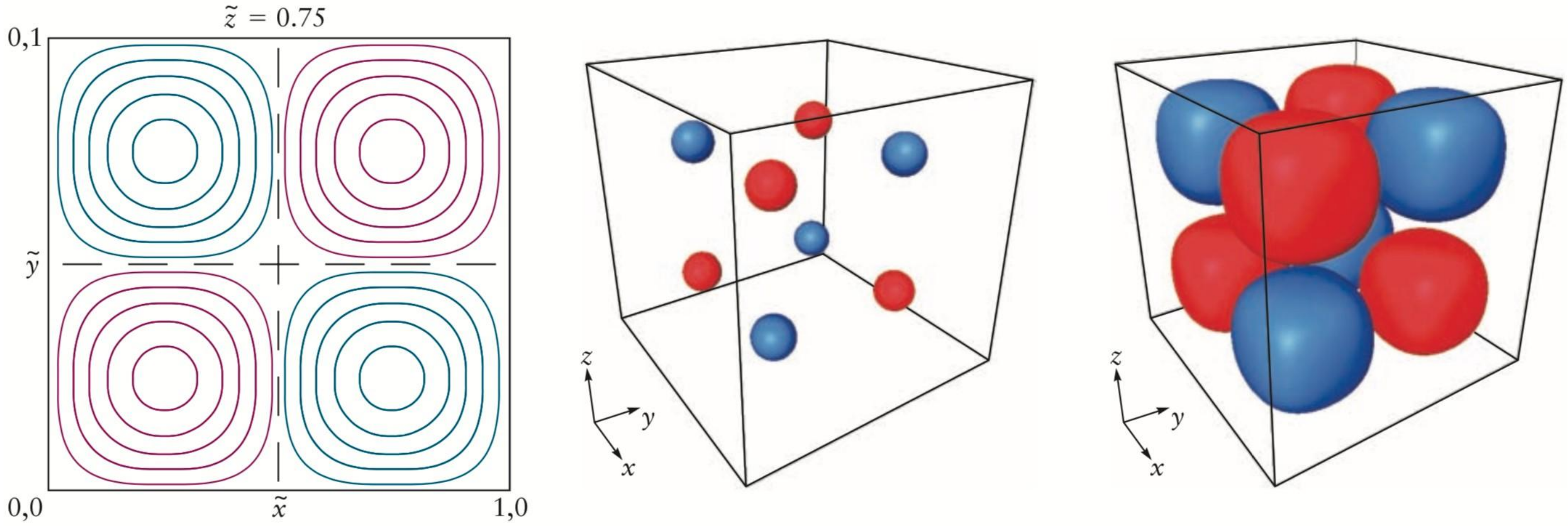


(a) (b)

(a) Isosurfaces for wave function value ψ_{123} = ± 0.8.

(b) Isosurfaces for wave function value ψ_{123} = ± 0.2.

Particle in cubic boxes



Representations of ψ_{222} (x, y, z) for a particle in a cubic box.

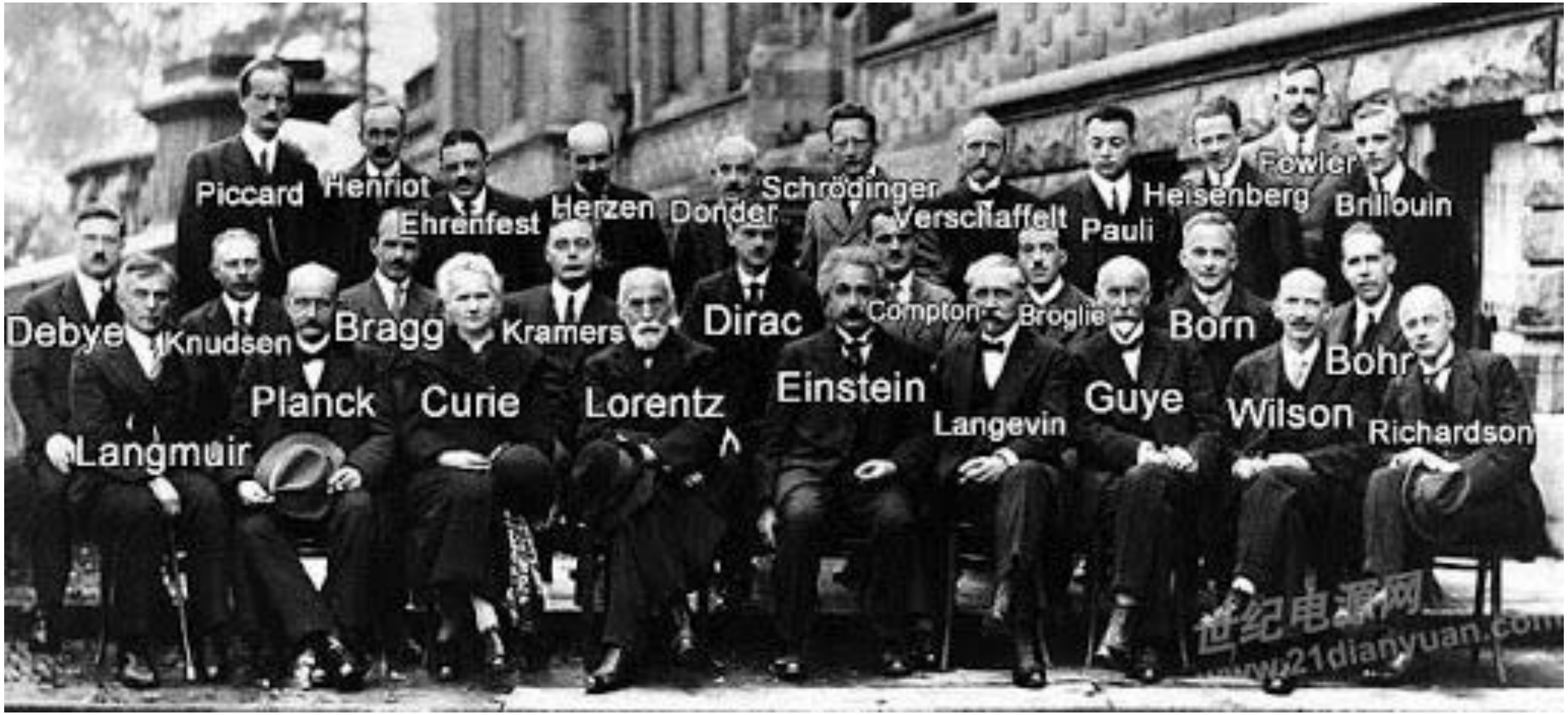
(a) Contour plots for a cut taken at z = 0.75.

(b) Isosurfaces for wave function value ψ_{222} = ± 0.9.

(c) Isosurfaces for wave function value ψ_{222} = ± 0.3.

Homework: read "principle of modern chemistry", chapter 4

"in search of Schrödinger's cat" by John Gribbin



"God does not play dice" Albert Einstein

"Don't tell god what to do" Neil Bohr